# Joint Process Monitoring in Power Plants Using Conditional Aggregate Mean Control Charts

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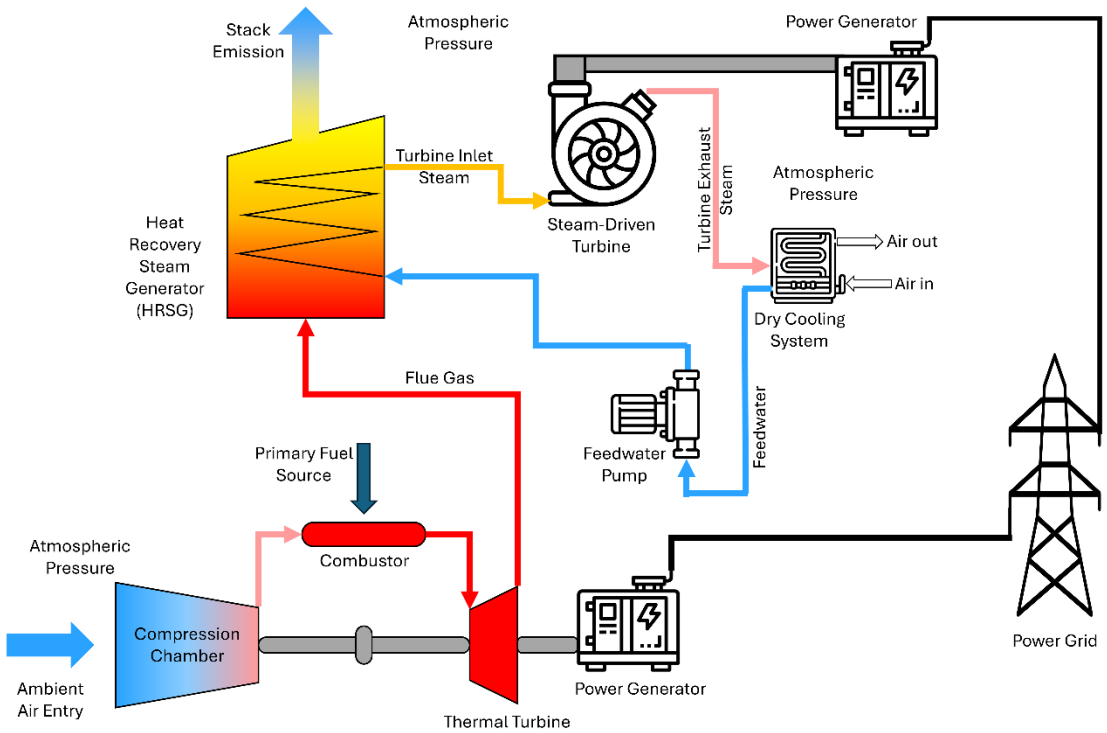
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## Abstract

Industrial processes are subject to inherent variations that require monitoring to distinguish between natural and unnatural fluctuations. Control charts are the most common tools used in statistical process control to monitor these changes. Traditionally, separate charts are used for monitoring process mean and variance. Recent studies favor joint charts that monitor both parameters, because they improve detection and diagnostic accuracy. We propose a new way to construct control charts based on the Conditional Aggregate Mean (CAM) approach for joint monitoring of process mean and variance. The method integrates three variance transformations, including first application of Johnson four parameter transformation within joint memory chart. We show that the proposed charts detect medium shifts faster than existing schemes. A case study on a power plant confirms their reliability and early detection of shifts. The charts suit power plant monitoring as they handle moderate and frequent parameter changes effectively.

**Keywords: Keywords:** Joint Monitoring; Statistical Process Control; Memory-Type Control Charts; CUSUM; EWMA; Variance Transformation; Combined Cycle Power Plant

## Graphical Abstract



# 1. Introduction

Ensuring process stability is a fundamental requirement across industrial and service sectors, as even minor deviations in operational parameters can lead to inefficiencies, increased costs, and compromised product or service quality. In manufacturing, fluctuations in machine performance can cause defects in mass production, leading to waste and rework. In healthcare, inconsistencies in medical procedures can impact patient safety, while in financial services, irregularities in transaction monitoring may indicate fraudulent activities. Similarly, in power generation, variations in process parameters such as turbine temperature, pressure, and fuel combustion efficiency can reduce energy efficiency, increase emissions, and cause premature equipment wear. These variations may originate from either natural fluctuation, which are inherent and do not significantly affect system performance, or assignable causes, which indicate unexpected disruptions that need immediate intervention. The ability to distinguish between these two types of variations is critical for optimizing industrial operations and ensuring sustainable production quality.

Among these industries, power plants, specifically Combined Cycle Power Plants (CCPPs), require rigorous process monitoring to maintain optimal performance. A CCPP integrates gas and steam turbines to enhance energy efficiency that is significantly higher than traditional single-cycle plants. However, due to the complexity of the system, CCPP performance is highly sensitive to shifts in key process variables, including atmospheric pressure, turbine efficiency, and heat exchange rates. Uncontrolled variations in these parameters can disrupt the combustion process, resulting in inefficient fuel consumption, fluctuating energy output, and potential damage to critical infrastructure. Monitoring and controlling these variations in real-time is essential to ensuring stable and efficient power generation. Given the scale and operational complexity of CCPPs, Statistical Process Control (SPC) techniques are widely employed to track and manage process stability [1].

SPC provides an analytical framework for distinguishing between normal process fluctuations and assignable variations that require corrective action. One of the most effective SPC tools is the control chart, first introduced by [2], which enables continuous process monitoring by visually displaying process behavior over time. Control charts allow engineers to detect shifts in key process parameters and identify potential causes before they result in costly inefficiencies or failures. A process is considered to be in an in-control (IC) state when it operates within expected limits, maintaining consistent output. However, when external disturbances or system failures introduce significant deviations in process mean or variability, the process transitions into an out-of-control (OOC) state, necessitating intervention. While traditional Shewhart control charts are effective in detecting large process shifts, they primarily monitor either the mean or variance separately, limiting their ability to detect simultaneous changes in both parameters. This is particularly problematic in CCPP operations, where variations in atmospheric pressure and turbine efficiency can affect both the location and dispersion of energy output [3]. For example, a sudden increase in the mean atmospheric pressure may indicate changes in external weather conditions affecting combustion efficiency, while an increase in variance could reflect instability in air intake or turbine control mechanisms. Thus, a comprehensive monitoring approach that can detect shifts in both parameters simultaneously is required to ensure reliable and efficient power generation.

To address these limitations, researchers have developed joint monitoring schemes that simultaneously track shifts in mean and variance within a single framework. Early joint monitoring approaches relied on independent control charts for mean and variance, but this method can be cumbersome and slow in detecting combined process deviations. More advanced techniques integrate both parameters into a unified charting structure, allowing for improved detection sensitivity, computational efficiency, and ease of interpretation. These advancements have proven highly beneficial in industries where real-time process control is essential, such as semiconductor manufacturing, chemical processing, and power generation [4,5].

Memory-type control charts, such as Cumulative Sum (CUSUM) and Exponentially Weighted Moving Average (EWMA) charts, have gained significant attention due to their ability to quickly detect small process shifts by incorporating past information into decision-making. Although these methods have improved sensitivity over traditional Shewhart charts, they often monitor mean and variance separately, making them less effective for identifying simultaneous changes. Recognizing these gaps, researchers have explored various joint monitoring techniques, including the Max-EWMA and Max-CUSUM schemes, which attempt to unify process mean and variance detection within a single framework. However, these approaches still face challenges in balancing sensitivity, interpretability, and computational complexity.

This study builds upon these advancements by introducing six memory-type control charts based on the Conditional Aggregate Mean (CAM) framework, specifically designed for joint monitoring of process mean and dispersion. The CAM-based approach integrates three different variance transformations, improving sensitivity of the charts to process shifts across various industrial applications. The proposed methodology is evaluated using Monte Carlo simulations, analyzing several key run-length (RL) performance metrics.

The CAM based charts are particularly suitable for CCPPs because the process involves interacting thermal and mechanical variables that change gradually over time. Memory type schemes are effective in such environments because they retain previous information, allowing detection of small or medium shifts that occur under varying load and weather conditions. The conditional aggregation mechanism may further help improve stability and reduce false alarms in systems where mean and variance change together, which reflects the actual operating behavior of combined cycle units. To illustrate the practical use of the CAM-based charts, a real-world case study using data from a CCPP is conducted.

Despite advancements in SPC and joint monitoring schemes, challenges remain in balancing sensitivity, interpretability, and computational efficiency. The present work contributes three distinct aspects. First, it applies the Johnson four parameter transformation within a joint memory type chart, which has not been previously considered for simultaneous monitoring of process mean and variance. Second, the proposed CAM charts are intended to detect shifts more quickly across a range of shifts compared with existing charts. Third, the charts are expected to maintain lower extra quadratic loss under both zero-state and steady-state settings, suggesting potential for consistent performance across process conditions. The scientific goal of this study is to improve joint monitoring theory by combining conditional aggregation with variance transformation to create a more sensitive framework for industrial process control. Because the performance of existing joint monitoring charts often depends

on specific parameter choices, the proposed use of the conditional aggregate mean within the joint structure aims to make the charts more robust across varying shift magnitudes without repeated parameter adjustment. This framework is expected to preserve sensitivity and reliability under diverse process conditions. Additionally, limited research has explored joint monitoring techniques tailored specifically for energy-intensive industries like CCPPs. This study addresses these challenges by introducing six memory-type control charts based on the CAM framework:

- The study developed six memory-type CAM-based control charts that simultaneously monitor process mean and variance.
- The efficiency of the proposed charts was evaluated by analyzing their run-length characteristics and comparing them with existing joint monitoring schemes such as Max-EWMA, Max-CUSUM, SS-EWMA, and SS-CUSUM charts.
- A case study on CCPP operations, where fluctuations in atmospheric pressure and turbine efficiency are critical, was used to validate the real-world applicability of CAM-based monitoring in power generation processes.

Based on these objectives, this study aims to answer the following research questions:

- How can memory-type control charts be improved to enhance joint monitoring of process mean and variance?
- How do CAM-based charts compare to existing joint monitoring methods in terms of detection speed and accuracy?
- What are the practical implications of applying CAM-based SPC monitoring in real-world CCPP operations?

The rest of the paper is structured as follows. Section 2 provides a comprehensive literature review of existing control charting methods and highlights the research gap in joint monitoring techniques. Section 3 contains the methodology, including variance transformations and the statistical formulation of CAM-based control charts. Section 4 presents the results of the study together with a detailed comparison between the proposed and existing schemes. Section 5 presents the practical application of the CAM charts using data from a CCPP process. Finally, Section 6 concludes the paper with the main findings and directions for future research.

## 2. Literature Review

SPC has been a widely used approach for monitoring and improving industrial processes by distinguishing between natural fluctuations and assignable variations. Since the introduction of control charts, researchers have continuously sought to enhance their sensitivity and applicability in complex manufacturing and process control environments. Traditional Shewhart-type control charts track either the mean or variance of a process independently. While effective in detecting large process deviations, these methods struggle with identifying smaller or gradual shifts, which can lead to unnoticed process deterioration. To overcome these limitations, memory-type control charts, such as CUSUM and EWMA charts, were introduced by [6] and [7], respectively. These methods incorporate cumulative and weighted past information to improve sensitivity to minor process variations [8]. Despite their

effectiveness, most conventional SPC techniques monitor only one process parameter, either the mean or variance, which limits their ability to detect simultaneous shifts in both.

To address this limitation, researchers have developed joint monitoring schemes that simultaneously track both mean and variance. [9] proposed one of the earliest methods for joint monitoring by combining two independent control charts into a single framework. [10] introduced a semicircle control chart that integrates both mean and variability within a unified monitoring approach. This was followed by the work of [11], who developed the Max chart, which integrates standardized mean and variance values using the maximum function to improve detection capabilities. A significant advancement in this direction was the Max-EWMA chart, introduced by [12], which combined two independent EWMA statistics, one for mean and another for variance, to enhance sensitivity to process deviations. [13] proposed the Max-Half-Mchart, an improved version of the Max-Mchart for simultaneously monitoring process mean and variability in individual observations. [14] introduced the variable sample size and sampling interval run sum Max chart, which dynamically adjusts sampling intervals to improve detection efficiency. These advancements reinforce the ongoing need for adaptable and computationally efficient SPC methods. Although these studies demonstrated the effectiveness of integrating location and dispersion information, many of these methods still required extensive parameter selection and had limitations in practical interpretability.

Further developments in EWMA-based joint monitoring techniques include the EWMA-Max and SS-EWMA charts [15], which introduced different ways of incorporating mean and variance information within an EWMA scheme. While these methods improved sensitivity to process deviations, they remained computationally complex. To enhance detection efficiency, [16] proposed the Max-DEWMA chart, which modified the Max-EWMA structure to better detect moderate shifts. [17] introduced a triple EWMA scheme, refining joint monitoring methods by incorporating a third smoothing parameter for greater adaptability. Additionally, [18] proposed the Max-P chart, which applies a progressive joint monitoring technique to improve performance under varying shift conditions. [19] extended joint monitoring methods with the Max-EWMAVAR and SS-EWMAVAR charts, which apply three-parameter transformations to enhance sensitivity in monitoring process dispersion alongside mean deviations. [20] developed a hybrid exponentially weighted moving average control chart, which employs a logarithmic transformation to normalize the distribution of sample variance. Their study highlighted that this transformation enhances variance tracking, improving the sensitivity of EWMA-based control charts for detecting both increases and decreases in process variability. Recently, [21] introduced a Max-EWMA scheme for monitoring shifted exponential distributions, further demonstrating the effectiveness of EWMA-based joint monitoring in industrial settings.

CUSUM-type joint monitoring schemes have also been explored extensively. [22] introduced the Max-CUSUM chart, which integrates two independent CUSUM statistics, one for mean and another for variance, into a single control chart. While this method showed improved responsiveness to process deviations, it required careful parameter selection. [23] proposed the SS-CUSUM chart, which combines squared CUSUM statistics of mean and variance to improve detection across various shift patterns. [24] introduced adaptive CUSUM approaches using unbiased estimators, providing improved shift detection across different process conditions. Similarly, [25] developed the maximum dual CUSUM chart, which enhances the detection of minor shifts in both parameters.

Beyond parametric SPC techniques, nonparametric joint monitoring methods have been developed to accommodate situations where the underlying process distribution is unknown or non-Gaussian. [26] introduced a nonparametric CUSUM-Lepage chart, which exhibited improved performance in detecting location and dispersion shifts without assuming normality. [27,28] proposed distribution-free Shewhart-Lepage-type control charts, which provide increased flexibility in diverse industrial applications. Recently, [29] introduced a nonparametric EWMA control chart, while [30] developed a hybrid Max-EWMA monitoring scheme that enhances sensitivity while reducing false alarm rates in non-Gaussian environments. [31] introduced a one-chart scheme for joint monitoring of zero-inflated Poisson processes, which shows the applicability of single monitoring frameworks in specialized distributions. Some advancements in SPC have also incorporated artificial intelligence (AI) and machine learning. [32] developed a convolutional neural network (CNN)-based EWMA chart to monitor spatially correlated processes, showcasing how AI-driven approaches can enhance SPC performance. [33] proposed deep learning models for forecasting energy output from a CCPP, highlighting how AI techniques can improve predictive maintenance and process control in power generation. While these approaches hold promise, their computational complexity remains a challenge for real-time industrial implementation.

Research in SPC has increasingly focused on applications in CCPPs, where effective monitoring is crucial to maintaining energy efficiency and reducing operational risks. [1] developed EWMA control charts based on the Reparametrized Birnbaum-Saunders regression model to monitor energy output in CCPPs, addressing the issue of asymmetric energy production datasets. [34] provided a theoretical discussion of CCPPs and their role in reducing emissions while maintaining high energy efficiency. [35] and [36] reviewed modern energy conversion technologies, emphasizing the importance of optimizing monitoring strategies for sustainable energy production. These studies highlight the increasing focus on improving energy efficiency and monitoring frameworks for better operational reliability in CCPPs.

Although significant progress has been made in joint SPC techniques, existing methods still exhibit limitations in adaptability, robustness, and sensitivity to varying process conditions. Furthermore, while recent studies (e.g., [1]) have explored monitoring techniques tailored for CCPPs, these methods either focus solely on location or dispersion parameters rather than providing an integrated solution for joint monitoring. This study intends to address this gap by developing an advanced framework that enhances joint monitoring capabilities, improves sensitivity to moderate shifts, and offers better practical applicability in CCPP operations. The subsequent sections will present the methodology, performance evaluation, and real-world application of the proposed CAM-based control charts.

# 3. Methodology

This section presents the mathematical formulation, transformation techniques, and construction of the CAM control charts proposed in this study. These charts are designed to simultaneously monitor process mean and dispersion, incorporating memory-based control mechanisms and variance transformations to improve sensitivity to process shifts.

## 3.1 Process Definition and Statistical Assumptions

Consider a sequence of independent observations $X_{1,t}, X_{2,t}, \ldots, X_{n,t}$ obtained from a normally distributed process $\{X_t; t \geq 1\}$ with a stable mean $\mu_0$ and standard deviation $\sigma_0$. The process is assumed to be IC for $t \leq \tau$, where $\tau$ represents the point at which a shift in the mean, variance, or both occurs, transitioning the process into an OOC condition. Mathematically, the process is defined as follows:

- when the process is IC, it follows $X_t \sim N(\mu_0, \sigma_0)$,
- when the process is OOC, it follows $X_t \sim N(\mu_1, \sigma_1)$, where $\mu_1 = \mu_0 + \delta\sigma_0$ and $\sigma_1 = \theta\sigma_0$,

where $\delta$ and $\theta$ represent the magnitude of shift in process mean and standard deviation, respectively. It should be noted here IC state corresponds to when $\delta = 0$ and $\theta = 1$. Four types of OOC scenarios are considered in this study: (N-D) represents a decrease in standard deviation, (U-N) represents an increase in mean, (U-D) represents an increase in mean along with a decrease in standard deviation, and (U-U) represents simultaneous increases in both mean and standard deviation. The study examines shifts that occur at the beginning of monitoring, referred to as zero-state conditions ($\tau = 1$), as well as shifts introduced after a period of IC operation, referred to as steady-state conditions ($\tau \to \infty$).

## 3.2 Transformations for Joint Monitoring

To jointly control the location and dispersion, we begin by defining the sample mean, $\bar{X}_t = (X_{1,t} + X_{2,t} + \cdots + X_{n,t})/n$, and sample variance, $S_t^2 = \sum_{i=1}^{n}\left(X_{i,t} - \bar{X}_t\right)^2/(n-1)$. Accordingly, $\bar{X}_t$ and $S_t^2$ are independent [37] and have the following distributions: $\bar{X}_t \sim N\left(\mu_0, \sigma_0/\sqrt{n}\right)$, and $(n-1)S_t^2/\sigma_0^2 \sim \chi_{n-1}^2$. To transform the mean and variance estimators ($\bar{X}_t$, $S_t^2$) to follow the same distribution, Cheng and Thaga [22] use the following transformations:

$$V_{1(t)} = \frac{\bar{X}_t - \mu_0}{\sigma_0/\sqrt{n}}, \tag{1}$$

$$V_{2(t)} = \varphi^{-1}\left[\omega_{\chi_{n-1}^2}\left(\frac{(n-1)S_t^2}{\sigma_0^2}\right)\right], \tag{2}$$

$\varphi^{-1}[.]$ here denotes the inverse of the cumulative distribution function (CDF) of a standard normal distribution and $\omega_{\chi_{n-1}^2}(.)$ represents the CDF of a Chi-squared distributed variable having degrees of freedom $(n-1)$. Under IC conditions, both $V_{1(t)}$ and $V_{2(t)}$ exhibit a standard normal distribution, $N(0,1)$. A change in the parameters of the process, however, appears in the mean and/or variance of the transformed variables, according to the nature of the deviation. Hence, new variables can be used to track irregular fluctuations in both parameters simultaneously.

To improve the detection capability of the joint charts, Ajadi and Riaz [38] suggested using a three-parameter logarithm transformation of $S_t^2$ before the normal standardization as follows:

$$T_{1,t} = a + b\ln(S_t^2 + c). \tag{3}$$

[39] defined the constants $a, b,$ and $c$ as follows: $b = B(n)$, $c = C(n)\sigma_0$, and $a = A(n) - 2B(n)\ln(\sigma_0)$,and showed that $T_{1,t}$ approximately follows normal distribution and has mean $E[T_{1,t}] = \mu_n$ and standard deviation $\sigma[T_{1,t}] = \sigma_n$. Where $\mu_n$ and $\sigma_n$ depend only on the sample size $n$. Using this information, we define another variable for controlling the variance as follows:

$$V_{3(t)} = \frac{T_{1,t} - \mu_n}{\sigma_n}. \tag{4}$$

It is well known that when sampling from the IC process, the distribution of $V_{3(t)}$ is also $N(0,1)$, and when the process is OOC, it is also reflected in the mean and/or variance of $V_{3(t)}$, depending on the nature of the deviation.

In an effort to make the distribution of $S_t^2$ more symmetric, [40] suggested transforming $S_t^2$ by applying a four-parameter Johnson transformation before using it in controlling the dispersion parameter. This transformation is expressed as follows:

$$T_{2,t} = a + b\left(\frac{S_t^2 - c}{c + d - S_t^2}\right). \tag{5}$$

The rationality of using this transformation and the distribution of $T_{2,t}$, as well as $E[T_{2,t}]$, $\sigma[T_{2,t}]$, and the constants $a, b, c$, and $d$ are given in Castagliola, Celano [40]. Based on $T_{2,t}$ we define another variable $V_{4,t}$ as follows:

$$V_{4(t)} = \frac{T_{2,t} - E[T_{2,t}]}{\sigma[T_{2,t}]}. \tag{6}$$

Similar to $V_{3(t)}$, $V_{4(t)} \sim N(0,1)$ when the process $\{X_t\}$ is IC and reflect changes in OOC situations. These transformed variables are used to construct the proposed CAM-based control charts.

### 3.3 Existing Control Charts for Joint Monitoring

To improve process monitoring efficiency, EWMA and CUSUM charts have been widely studied for detecting small to moderate shifts in process parameters. These methods have been extended for joint monitoring of mean and variance using two combination techniques: Max and SS. The Max approach takes the maximum of absolute monitoring statistics, while the SS approach sums their squared values.

The Max-EWMA control chart employs two independent EWMA statistics, one for detecting changes in the mean and another for the variance. These are defined as:

$$\left.\begin{aligned} E_{1(t)} &= \lambda_1 V_{1(t)} + (1 - \lambda_1) E_{1(t-1)} \\ E_{2(t)} &= \lambda_1 V_{2(t)} + (1 - \lambda_1) E_{2(t-1)} \end{aligned}\right\}, \tag{7}$$

where $0 < \lambda_1 \leq 1$ is the smoothing parameter that determines the weight of current sample. The overall monitoring statistic for the Max-EWMA chart is given by $M_{1(t)} = \max\left(\left|E_{1(t)}\right|, \left|E_{2(t)}\right|\right)$ having an upper control limit (UCL): $\sqrt{\lambda_1/(2 - \lambda_1)}\,(1.128379 + 0.602810L_1)$. Here, $L_1$ controls the width of the limit.

In a similar way, [38] proposed another version of the EWMA chart by replacing the variable controlling the dispersion parameter in equation (7) with the transformation defined in equation (4) to form a new chart called EWMAVAR (denoted EWMA-3T in this study). Similarly, Xie [15] proposed combining the EWMA statistics defined in equation (7) by the method of sum of squares to construct a new chart called SS-EWMA (SS-EWMA-$\chi^2$) To concurrently monitor the mean and variability using one charting statistic, $SS_{1(t)} = E_{1(t)}^2 + E_{2(t)}^2$, having a mean, $E\left(SS_{1(t)}\right) = 2\sigma_E^2$, and variance, $\sigma^2\left(SS_{1(t)}\right) = 4\sigma_E^4$, where $\sigma_E^2 = \sigma_{E_{1(t)}}^2 = \sigma_{E_{2(t)}}^2$. The asymptotic UCL for the SS-EWMA-$\chi^2$ chart is given as $(2\lambda/(2-\lambda))(1 + L_2)$, where $L_2$ is the coefficient that controls the width of the limit. The SS-EWMAVAR (SS-EWMA-3T) of Ajadi and Riaz [38] is formed in a similar way by replacing $V_{2(t)}$ in equation (7) with $V_{3(t)}$ from equation (4).

The Max-CUSUM control chart tracks shift in mean and variance using two independent, two-sided CUSUM statistics:

$$\left.\begin{aligned} MC_{1(t)}^{+} &= \max\left(0, MC_{1(t-1)}^{+} + V_{1(t)} - k_1\right) \\ MC_{1(t)}^{-} &= \max\left(0, MC_{1(t-1)}^{-} - V_{1(t)} - k_1\right) \end{aligned}\right\}, \tag{8}$$

$$\left.\begin{aligned} MC_{2(t)}^{+} &= \max\left(0, MC_{2(t-1)}^{+} + V_{2(t)} - k_1\right) \\ MC_{2(t)}^{-} &= \max\left(0, MC_{2(t-1)}^{-} - V_{2(t)} - k_1\right) \end{aligned}\right\}, \tag{9}$$

where $k_1$ is the reference parameter. A single plotting statistic is developed by combining these values for the Max-CUSUM (Max-CUSUM-$\chi^2$) chart, $M_{2(t)} = \max\left(MC_{1(t)}^{+}, MC_{1(t)}^{-}, MC_{2(t)}^{+}, MC_{2(t)}^{-}\right)$. The initial values of the statistics in equations (8) and (9) are set to zero and $k_1$ is reference parameter of the CUSUM chart that controls its sensitivity to changes in the parameter(s). An OOC signal is given when $M_{2(t)} > h_1$, where $h_1$ is the threshold limit. In a related manner [38] created the Max-CUSUMVAR (Max-CUSUM-3T) by substituting $V_{2(t)}$ in equation (8) and (9) with $V_{3(t)}$ from equation (4). Further enhancement was achieved by [23] with the SS-CUSUM chart, where the sum of squares of the maximum CUSUM statistics from equations (8) and (9) was calculated as follows:

$$S_{1(t)} = \max\left(MC_{1(t)}^{+}, MC_{1(t)}^{-}\right),$$

$$S_{2(t)} = \max\left(MC_{2(t)}^{+}, MC_{2(t)}^{-}\right),$$

$$SS_{2(t)} = S_{1(t)}^2 + S_{2(t)}^2.$$

The SS-CUSUM (SS-CUSUM-$\chi^2$) chart is then constructed by plotting $SS_{2(t)}$ against a specified threshold, $h_2$. [38] created a new chart, known as the SS-CUSUMVAR (SS-CUSUM-3T), by modifying this approach: substituting $V_{2(t)}$ in equation (8) and (9) with $V_{3(t)}$ from equation (4) to incorporate the three-parameter transformation, enabling more refined joint monitoring of process variability alongside mean changes. The four-parameter Johnson transformation in equation (5) has not, to the best of our knowledge, been applied in prior studies for joint monitoring, and therefore it can be considered an additional contribution of the current work.

### 3.4 Proposed Control Charts for Joint Monitoring

Building on these foundations, this study introduces six memory-type CAM-based control charts designed to improve detection capabilities by incorporating cumulative aggregation mechanisms. The first set of charts, referred to as Max-CAM charts, includes Max-CAM-$\chi^2$, Max-CAM-3T, and Max-CAM-4T, which utilize Chi-square, three-parameter logarithm, and four-parameter Johnson transformations, respectively. The second set of charts, referred to as SS-CAM charts, includes SS-CAM-$\chi^2$, SS-CAM-3T, and SS-CAM-4T, which apply the sum of squares approach to the same transformations. The main difference between the existing schemes, such as CUSUM and EWMA, and our proposed approach is that the former use fixed recursive weighting or cumulative sums, while the latter applies conditional averaging that adjusts to recent process behavior. This mechanism allows the chart to capture the joint effect of location and scale changes more efficiently.

The Max-CAM charts are constructed by first defining two independent two-sided CUSUM statistics for detecting shifts in mean and variance. These statistics are defined as:

$$\left.\begin{aligned} CUSUM^{+}_{1(t)} &= \max\left(0, CUSUM^{+}_{1(t-1)} + V_{1(t)} - k_2\right) \\ CUSUM^{-}_{1(t)} &= \max\left(0, CUSUM^{-}_{1(t-1)} - V_{1(t)} - k_2\right) \end{aligned}\right\}, \tag{10}$$

$$\left.\begin{aligned} CUSUM^{+}_{2(t)} &= \max\left(0, CUSUM^{+}_{2(t-1)} + V_{2(t)} - k_2\right) \\ CUSUM^{-}_{2(t)} &= \max\left(0, CUSUM^{-}_{2(t-1)} - V_{2(t)} - k_2\right) \end{aligned}\right\}, \tag{11}$$

where $k_2$ is the reference parameter. The starting values are set as $CUSUM^{+}_{1(0)} = CUSUM^{-}_{1(0)} = CUSUM^{+}_{2(0)} = CUSUM^{-}_{2(0)} = 0$.

Next, we introduce two pairs of discrete variables, $K^{+}_{1(t)}$and $K^{-}_{1(t)}$, along with $K^{+}_{2(t)}$and $K^{-}_{(2)t}$, which respectively count the number of samples examined since the last occurrences of $CUSUM^{+}_{1,t} = 0$, $CUSUM^{-}_{1,t} = 0$, $CUSUM^{+}_{2,t} = 0$, and $CUSUM^{-}_{2,t} = 0$, as defined below:

$$K^{+}_{1(t)} = \begin{cases} 1 & \text{if } CUSUM^{+}_{1(t-1)} = 0 \\ 1 + K^{+}_{1(t-1)}, & \text{otherwise} \end{cases}, \quad K^{-}_{1(t)} = \begin{cases} 1 & \text{if } CUSUM^{-}_{1(t-1)} = 0 \\ 1 + K^{-}_{1(t-1)}, & \text{otherwise} \end{cases}, \tag{12}$$

$$K^{+}_{2(t)} = \begin{cases} 1 & \text{if } CUSUM^{+}_{2(t-1)} = 0 \\ 1 + K^{+}_{2(t-1)}, & \text{otherwise} \end{cases}, \quad K^{-}_{2(t)} = \begin{cases} 1 & \text{if } CUSUM^{-}_{2(t-1)} = 0 \\ 1 + K^{-}_{2(t-1)}, & \text{otherwise} \end{cases}, \tag{13}$$

In equations (12) and (13), the definitions of the discrete variables ensure that the count for $K^{+}_{1(t)}$, $K^{-}_{1(t)}$, $K^{+}_{2(t)}$, $K^{-}_{2(t)}$ restarts at 1 whenever the respective previous CUSUM statistic $CUSUM^{+}_{1(t-1)}$, $CUSUM^{-}_{1(t-1)}$, $CUSUM^{+}_{2(t-1)}$, or $CUSUM^{-}_{2(t-1)}$ resets to zero. This resetting mechanism effectively eliminates the drawback in detecting delayed shifts often faced by aggregate mean charts. The proposed Max-CAM-$\chi^2$ chart incorporates these variables in the following statistics to track increases and decreases in the mean (equation 14) and variance (equation 15) of a normal process:

$$\left.\begin{aligned} CAM^{+}_{1(t)} &= \left(1 - \frac{1}{K^{+}_{1(t)}}\right) CAM^{+}_{1(t-1)} + \frac{1}{K^{+}_{1(t)}} V_{1(t)} \\ CAM^{-}_{1(t)} &= \left(1 - \frac{1}{K^{-}_{1(t)}}\right) CAM^{-}_{1(t-1)} + \frac{1}{K^{-}_{1(t)}} V_{1(t)} \end{aligned}\right\}, \tag{14}$$

$$\left.\begin{aligned} CAM^{+}_{2(t)} &= \left(1 - \frac{1}{K^{+}_{2(t)}}\right) CAM^{+}_{2(t-1)} + \frac{1}{K^{+}_{2(t)}} V_{2(t)} \\ CAM^{-}_{2(t)} &= \left(1 - \frac{1}{K^{-}_{2(t)}}\right) CAM^{-}_{2(t-1)} + \frac{1}{K^{-}_{2(t)}} V_{2(t)} \end{aligned}\right\}. \quad (15)$$

Since the CAM statistics in equations (14) and (15) represent the cumulative average of the transformed variables $V_{1(t)}$and $V_{2(t)}$ from the point when the CUSUM statistics were last reset to zero, the mean values of the statistics under IC state are (see detailed proof in Appendix A) $E\left(CAM^{+}_{1(t)}\right) = E\left(CAM^{-}_{1(t)}\right) = E\left(CAM^{+}_{2(t)}\right) = E\left(CAM^{-}_{2(t)}\right) = 0$, and the and sampling variances are given as, $\mathrm{Var}\left(CAM^{+}_{1(t)}\right) = 1/K^{+}_{1(t)}$, $\mathrm{Var}\left(CAM^{-}_{1(t)}\right) = 1/K^{-}_{1(t)}$, $\mathrm{Var}\left(CAM^{+}_{2(t)}\right) = 1/K^{+}_{2(t)}$, and $\mathrm{Var}\left(CAM^{-}_{2(t)}\right) = 1/K^{-}_{2(t)}$. The CAM statistics are incorporated into the Max statistic, which takes the highest value of the absolute scaled cumulative averages and serves as the plotting statistic for the proposed charting scheme. The independence of $CAM^{+}_{1(t)}$, $CAM^{-}_{1(t)}, CAM^{+}_{2(t)}$, and $CAM^{+}_{2(t)}$ is implied by the fact that $V_{1(t)}$ and $V_{2(t)}$ are independent. Mathematically, the MaxCAM statistic is defined as:

$$\mathrm{Max}CAM_{(t)} = \max\left(\left|CAM^{+}_{1(t)} \times \sqrt{K^{+}_{1(t)}}\right|, \left|CAM^{-}_{1(t)} \times \sqrt{K^{-}_{1(t)}}\right|, \left|CAM^{+}_{2(t)} \times \sqrt{K^{+}_{2(t)}}\right|, \left|CAM^{-}_{2(t)} \times \sqrt{K^{-}_{2(t)}}\right|\right) \quad (16)$$

To construct the proposed Max-CAM-$\chi^2$ chart, $\mathrm{Max}CAM_t$ is calculated at each time point $t = 1,2,..$ and plotted to monitor process stability. $\mathrm{Max}CAM_t$ overtime reveals whether the process remains stable:

- A large $\mathrm{Max}CAM_{(t)}$ value indicates that the process mean, the variance, or both have likely shifted from their IC levels, and such a condition is confirmed when the statistic exceeds the control limit threshold.
- A smaller $\mathrm{Max}CAM_{(t)}$ suggests that both the mean and variance are near their target levels.

Since $\mathrm{Max}CAM_{(t)}$ is non-negative, it is compared only to a specified positive threshold, say $h_2$. Values of $\mathrm{Max}CAM_{(t)}$ above $h_2$ signal a potential process shift, while values below it indicate the process is likely stable.

The two additional Max charts designed to simultaneously track changes in both mean and variance, Max-CAM-3T and Max-CAM-4T, are based on three- and four-parameter transformations of the sample variance. They are constructed in a similar manner by substituting $V_{2(t)}$ in equations (11) and (15) with $V_{3(t)}$ from equation (4) for Max-CAM-3T chart and $V_{4(t)}$ from equation (6) for Max-CAM-4T chart.

The SS-CAM charts apply a sum of squares approach, combining the squared CUSUM statistics into a single monitoring statistic. Starting with SS-CAM-$\chi^2$, we define two variables that take the maximum of the scaled CAM statistics from equations (17) and (18) as follows:

$$CAM_{1(t)} = \max\left(\left|CAM^{+}_{1(t)} \times \sqrt{K^{+}_{1(t)}}\right|, \left|CAM^{-}_{1(t)} \times \sqrt{K^{-}_{1(t)}}\right|\right), \quad (17)$$

$$CAM_{2(t)} = \max\left(\left|CAM^{+}_{2(t)} \times \sqrt{K^{+}_{2(t)}}\right|, \left|CAM^{-}_{2(t)} \times \sqrt{K^{-}_{2(t)}}\right|\right). \quad (18)$$

Using to these statistics, the new chart is constructed by computing $SSCAM_{(t)} = CAM^2_{1(t)} + CAM^2_{2(t)}$ and plotting it against a predetermined threshold, say $h_3$. An OOC scenario is declared at time $t$ if $SSCAM_{(t)} > h_3$. Like the Max-CAM charts, the SS-CAM-3T and SS-CAM-4T charts for joint monitoring are based on three and four-parameter transformations.

## 3.5 Performance Metrics

This study uses the average run length (ARL), defined as the expected count of samples investigated until an OOC signal is generated, in evaluating the efficiency of the new charts and comparing them to other existing options. In the case of an IC process, the ARL is expressed as $ARL_0$, while for OOC situations, it is indicated as $ARL_1$. To provide a comprehensive assessment of chart efficiency, we have also reported the median run length (MDRL) and the standard deviation of run length (SDRL). The median offers greater robustness when the RL distribution is skewed, and SDRL measures the variation in the RL, where a reduced SDRL indicates more consistent performance of the chart. The conditional expected delay (CED) curve is another important individual performance metric which assesses the efficiency of the control chart across different change points $\tau$. Specifically, $CED_\tau$ represents the expected run length at $\tau$ given that no alarms were triggered prior to this change point.

Additionally, overall performance is usually evaluated using extra quadratic loss (EQL), relative average run length (RARL), and the performance comparison index (PCI). These metrics aid in evaluating the effectiveness of the charts in a range of shifts. Refer to [41] for more discussion on these measures. This study incorporated the EQL metric estimated using the following formula:

$$EQL = \frac{1}{M}\sum_{i=1}^{M}(\delta_i^2 + \theta_i^2)ARL(\delta_i\theta_i)$$

In this context, $M$ represents the number of OOC observations considered and $ARL(\delta_i, \theta_i)$ is the $i$th $ARL_1$ value for the joint shift $\delta_i, \theta_i$.

The measures used in this study are drawn from prior SPC literature and follow the definitions used by [41] and similar studies on control charts. They provide a consistent way to evaluate detection speed, stability, and false alarm behavior.

### 3.6 Research Design and Simulation

The literature offers several ways to compute the RL for control charts, including Monte Carlo simulations, integral equations, and Markov Chain methods. The experimental design uses Monte Carlo simulation to estimate RL characteristics for all charts because the other methods do not work for all chart types, especially memory-type control charts, where the nature of the plotting statistics can make other methods less practical. The ARL values of the

proposed charts are computed using Algorithm 1. The overall process is illustrated in Figure 1, which shows the flow of sampling, computations, and the iteration steps.

The main parameters are as follows, reported to allow replication and explicit interpretation. The base sample size is $n$ = 5. The reference parameter $k_1$ is set to 0.5 for CUSUM based schemes while EWMA chart uses lambda equal to 0.2 and the IC ARL ($ARL_0$) is fixed at 250, consistent with prior work [12,15,17,42]. For each design, the simulation uses $m = 10^6$ Monte Carlo replications to estimate ARL and related measures. The reported ARL values are the mean of the $m$ run lengths. The control limits $h$ are found by iterative search. No fixed random seed is applied in these experiments to avoid inadvertent dependence on a single sequence of pseudo random numbers. The set of OOC shifts includes $\delta$ values in the set (0.2, 0.5, 0.75, 1.0, 1.25, 1.5, 2.0) and $\theta$ values in the set (0.75, 0.9, 1.1, 1.25, 1.5, 2.0). Change points (time points where the change occurs) $\tau$ are examined over the range 1 to 50 for steady state comparisons. The selection of these values follows standard choices in the SPC literature and permits comparison with existing joint scheme results as indicated.

Algorithm 1: ARL computation of the proposed charts

1. Fix the sample size $n$ and desired $ARL_0$, and select the reference parameter $k_2$, limit threshold $h$, and change point $\tau. Set\ \tau = 1$ for zero-state.
2. At each time $t$, draw $n$ random observations according to the following conditions:
   *i.* $X_t \sim N(\mu_0, \sigma_0)$, if $t \leq \tau$.
   *ii.* $X_t \sim N(\mu_1 = \mu_0 + \delta\sigma_0, \sigma_1 = \theta\sigma_0)$, otherwise.
3. Compute $\bar{X}_t$and $S_t^2$ and apply the chosen transformation.
4. Compute the CUSUM statistics mean and variance.
5. Compute the $K$ variables from equations 13 and 14.
6. Compute the CAM statistics.
7. Combine the statistics in step #6 into a single statistic using either the $\max CAM_{(t)}$ defined for Max-CAM charts, or $SSCAM_{(t)}$ for SS-CAM charts.
8. If the statistic exceeds $h$ at time $t$, record $RL = t - \tau + 1$ . Otherwise go to Step 2.
9. Repeat Steps 2 to 8 for $m$ runs, for example $10^6$ runs.
10. Compute $ARL$ as the mean of the $m$ $RL$s as.
11. For IC process, if $ARL_0$ is not close to the target, adjust h and repeat.
12. For $ARL_1$, repeat Step 2 with $\delta$ not equal to 0, or $\theta$ not equal to 1, or both.

**[Insert Figure 1 here]**

## 4. Results

The performance of the proposed CAM-based control charts is evaluated in this section by analyzing their RL characteristics and comparing them with existing joint monitoring schemes. The effectiveness of these charts is assessed under different shift scenarios, considering both zero-state and steady-state conditions.

## 4.1 Performance Evaluation

This study assessed the ARL of the proposed CAM charts with a sample size of $n = 5$, varying sample sizes can be explored to understand their impact. Throughout the study, we set the reference parameter $k_2$ to 0.5. Moreover, additional values of $k_2$ are provided in Tables B2 – B4 to capture its effect on the performance of the charts. Additionally, to facilitate reproducibility of the results in this study, we have provided limits thresholds to achieve $ARL_0 \cong 250$ for all the proposed charts in Table B1.

Table 1 presents the ARL values of the proposed charts together with SDRL and MDRL for a range of deviations in the mean $\left(\delta \in (0,0.1,0.2,0.3,0.4,0.5,0.75,1,1.25,1.5,2,3)\right)$ and variance $\left(\theta \in (0.25,0.5,0.75,0.9,1,1.1,1.25,1.5,2)\right)$ of a process. The results show that the proposed charts are highly sensitive to changes in the parameters of the process. For instance, the ARL values for the Max-CAM-3T chart reduced from 250 for IC state to 3.53 when a shift $\theta = 0.25$ is introduced in the process variance. Comparable reductions are observed across the other charts, with the Max-CAM-$\chi^2$ having the least ARL of 2.31.

All charts yield relatively high ARL values when only the process variance experiences a small change (e.g., $\theta = 0.9$). However, the ARLs decrease significantly when the shift is accompanied by even a small change in the mean. For instance, with a joint shift of $(\delta, \theta) = (0.2,0.9)$ the ARL for the SS-CAM-4T chart decreases by approximately 80%, from 250 to as low as 47.88.

In terms of the MDRL, all the charts show almost similar behavior across different shift combinations. In almost all the cases, $\mathrm{ARL} > \mathrm{MDRL}$ indicating a positive skewness in the RL distribution. For example, when $\delta = 0.2$ and $\theta = 1.25$, the Max-CAM-3T has $\mathrm{ARL} = 13.33$ and $\mathrm{MDRL} = 11$, and SS-CAM-3T has 12.39 and 10 respectively

The SDRL values are also observed to decline as the shifts become larger, indicating that the charts perform more consistently when identifying medium to substantial shifts in the parameters.

It can noted in general, the charts that use the three-parameter transformation (Max-CAM-3T and SS-CAM-3T) have better performance than the others when there is moderate change in the process variability. Moreover, for $\delta \geq 0.5$ and $\theta$ is moderate to large, the performance of all six charts is nearly the same.

**[Insert Table 1 here]**

## 4.2 Comparative Analysis

The results from the previous section show the effectiveness of the proposed CAM-based control charts in detecting shifts in both process mean and variance. This section presents a comparative evaluation of the proposed and existing EWMA and CUSUM charts.

To ensure a fair comparison, the design parameters for the competing schemes are set as follows: for all proposed charts and CUSUM charts, $k = 0.5$; for all EWMA charts, $\lambda = 0.2$. Additionally, $ARL_0$ is fixed at 250, and the sample size is set to $n = 5$.

Table 2 presents a detailed (ARL, SDRL, MDRL) comparison among the proposed and CUSUM and EWMA charts, based on two methods (SS and Max) and three transformations (3T, 4T, and Chi-squared). Various shifts have been considered to compare the charts effectiveness in identifying changes in both parameters of a normal process. To be concise, we have considered the following scenarios: minor and medium shifts in the process location only ($\delta = 0.2, 0.5, \theta = 1$); minor and medium shifts in the process scale only ($\delta = 0, \theta = 0.75, 1.5$); and hybrid and major shifts in both parameters ($\delta = 0.2,1.5, \theta = 0.75,1.5$).

The comparison of the proposed charts with existing control chart methods aims to show how the proposed CAM structure performs relative to approaches commonly available in the literature. Evaluating multiple methods under the same process conditions provides clear evidence of which structure responds best to shifts in both mean and variance.

From Table 2, the following observations can be summarized:

- ***Max-CAM charts vs Max-CUSUM charts:*** The proposed Max-CAM charts have advantage over the Max-CUSUM charts in identifying medium changes in the process level and/or scale. For instance, at $\delta = 1.5 \text{ and } \theta = 1.0$, the ARL for the Max-CAM-3T is reduced by $99.35\%$, reaching 1.62, compared to the $99.07\%$ reduction for the Max-CUSUM-3T, which has an ARL of 2.32.
- ***Max-CAM charts vs Max-EWMA charts:*** In most cases, the Max-CAM charts consistently dominate the Max-EWMA charts when identifying both small and medium changes, as indicated by their ARL values. The only exception is when $\delta = 0.2 \text{ and } \theta = 1$, where both charts show similar performance.
- ***SS-CAM charts vs. SS-CUSUM charts:*** Similar to the Max-CAM the SS-CAM charts demonstrate a some advantage over SS-CUSUM charts in identifying medium changes in the process mean and variance. For instance, under the $\chi^2$ transformation with $\delta = 1.5$ and $\theta = 1.5$, the ARL for SS-CAM decreases by $99.32\%$, reaching 1.71, while SS-CUSUM shows a $99.08\%$ reduction, with an ARL of 2.31. This advantage is evident for $\delta = 1.5$ and $\theta = 1.5$. However, for smaller shifts, the SS-CUSUM charts perform better than SS-CAM, though the differences are marginal.
- ***SS-CAM charts vs SS-EWMA charts:*** The SS-CAM charts uniformly outperform the SS-EWMA charts especially when identifying medium changes in the parameters. However, for small shifts the SS-EWMA charts take the lead. As an example, the ARL for the SS-EWMA-$\chi^2$ is 18.95 when $\delta = 0.2$ and $\theta = 0.75$, while that of SS-CAM-$\chi^2$ is 20.55, indicating a $92.42\%$ and $91.78\%$ ARL reduction for SS-EWMA-$\chi^2$ and SS-CAM-$\chi^2$ respectively.

The proposed Max-CAM and SS-CAM schemes perform well, in many instances superior to the Max-CUSUM and SS-CUSUM schemes, especially when there are moderate changes in the process mean and/or variability. The CAM-based schemes detect shifts faster in most situations, as indicated by low ARL values when larger shifts occur. The charts mostly dominate the Max-EWMA charts. Similarly, the SS-CAM charts generally perform better than SS-EWMA for moderate shifts, though SS-EWMA has a slight advantage with smaller shifts.

Overall, CAM-based charts provide a balanced approach to detecting shifts in process parameters, making them especially useful for keeping processes stable when moderate shifts are important.

Because the exact shift in process parameters is usually not known beforehand, we have also considered the overall efficiency of the charts across various shifts. In addition to the ARL values, the EQL metric is also reported in Table 2. As expected, all the proposed charts have lower EQL values compared to their CUSUM and EWMA counterparts. This is an indication of the superiority of our proposals in detecting shifts efficiently. The lower EQL values demonstrate the ability of the charts to maintain a quicker and more consistent response across a wide range of shifts rather than specific shifts.

**[Insert Table 2 here]**

The steady-state performance, in most cases, is considered more relevant than the zero-state because it better reflects real-life scenarios. Under the zero-state condition, the process is assumed to go OOC from the beginning, but in many instances the process remains IC for some time before a change in the process parameters occurs, which aligns with the steady-state assumption. To this end, we have analyzed the IC and OOC CED curves of the proposed charts using the same OOC shifts considered in the zero-sate and compared them with other competing charts at various change-points $\tau \in [1,50]$. The results are presented as plots in Figures 2 – 6.

**[Insert Figures 2 – 6 here]**

In the IC state (Figure 2), all the charts show a reduction in $ARL_0$, indicating an increased false alarm rate in the steady-state. The CUSUM charts are the most impacted, and next are the proposed charts, with the EWMA charts being the least impacted. In general, SS-based schemes are more affected than their Max counterparts.

Figures 3 and 4 compare the OOC CED curves of the proposed CAM schemes with those of the CUSUM schemes for different minor to medium shifts, while Figures 5 and 6 show the comparison with the EWMA charts. In the OOC state, all charts exhibit a slight reduction in $ARL_1$ as $\tau$ increases, reflecting good steady-state performance.

The proposed charts are shown to be superior to their CUSUM counterparts in identifying medium shifts. Specifically, when $\delta = 1.5$ and/or $\theta = 1.5$, the CUSUM charts are less effective compared to the proposed charts. However, for smaller shifts, such as $\delta = 0.2$ and $\theta = 0.75$, the CUSUM charts perform better. The CED comparisons with EWMA schemes in Figures 5 and 6 show that the CAM-based schemes generally outperform their EWMA alternatives, with consistently lower CED values, except in cases of small shifts.

For practical applications, quality and process control specialists should review the ARL, CED, and EQL values and describe their preferences in simple linguistic terms such as better, comparable, or lower.

## 5. Case Study on Process Monitoring in a CCPP

To illustrate the practical applicability of the proposed CAM control charts, this section presents a real-world case study using operational data from a CCPP. The objective is to highlight the importance of SPC in power generation and illustrate how the proposed charts enhance monitoring efficiency.

A CCPP is an advanced power generation system that integrates two thermodynamic cycles, i.e. the Brayton cycle, which operates on a gas turbine, and the Rankine cycle, which utilizes a steam turbine. The gas turbine burns natural gas, producing high-temperature exhaust gases that drive a turbine to generate electricity. Instead of discarding this waste heat, it is captured and used to produce steam, which then powers a steam turbine, generating additional electricity. Maintaining optimal operational conditions in a CCPP is essential for maximizing fuel efficiency, minimizing emissions, and ensuring the longevity of critical equipment. Any unexpected deviations in key process variables can lead to decreased efficiency, increased operational costs, and potential safety hazards. Monitoring these variations in real-time is crucial for maintaining a stable and efficient power generation process. Several process variables influence the efficiency and stability of a CCPP, including vacuum pressure, atmospheric pressure, relative humidity, and ambient temperature. Among these, atmospheric pressure (AP) is particularly critical because it directly affects air intake, combustion efficiency, and heat exchange processes. Figure 7 provides a schematic representation of the atmospheric pressure variations in a CCPP, highlighting key areas where AP plays a significant role in power generation efficiency. Fluctuations in atmospheric pressure can lead to variations in power output, inefficient fuel combustion, and increased mechanical stress on turbine components. Monitoring only the mean of AP is insufficient because a process may appear stable in terms of its central value while experiencing significant fluctuations in variability. A sudden increase in AP variance, even if the mean remains within acceptable limits, could indicate irregularities in combustion efficiency, leading to inconsistent power output. Conversely, a decrease in variance might suggest an overly controlled process that could be masking underlying inefficiencies. Simultaneously monitoring both mean and variance ensures that process deviations are identified promptly, allowing corrective actions to be taken before they affect overall plant performance.

**[Insert Figure 7 here]**

Before implementing the control charts, it is essential to verify that the data satisfies key statistical assumptions, particularly normality and independence. The first 600 atmospheric pressure measurements were analyzed as IC data to estimate the baseline process parameters, with the mean found to be 1013.55 and the standard deviation 5.82. The Anderson-Darling test is applied to assess normality, yielding a $p$-value of 0.4765, indicating that the AP values are not significantly different from a normal distribution. To verify the independence of observations, the Ljung-Box test is conducted, producing a $p$-value of 0.7782. A high p-value suggests no significant autocorrelation within the data, supporting the assumption that the observations are independent over time. These results confirm that the dataset meets the fundamental assumptions required for control chart implementation.

To monitor atmospheric pressure variations, the proposed CAM control charts are implemented alongside traditional CUSUM and EWMA charts for comparison. The first step in chart design is the selection of charting constants that balance sensitivity to small shifts while minimizing false alarms. For the CAM and CUSUM charts, the reference parameters are

set as $k_1 = 0.5$ and $k_2 = 0.5$, ensuring an appropriate balance between sensitivity and stability. The control limits were established at $h_1 = 5.05$ and $h_2 = 3.453$ to maintain an $ARL_0$ of approximately 250. For the EWMA chart, a smoothing parameter of $\lambda_1 = 0.2$ is selected, with a limit coefficient $L_1 = 3.036$.

Figure 8 shows the implementation of the control charts. The first 20 samples are plotted against a green background, indicating that the process was IC during this period. After sample number 20 (red background) a shift in the process mean and/or variance has occurred. In the first scenario ($\delta$ = 0.2, $\theta$ = 0.75), the CAM and EWMA charts are the first to detect an OOC condition, triggering an alarm at sample 45. The CUSUM chart signaled an OOC state at sample 48, showing a slightly delayed response. In total the CAM chart detected six OOC points, while the CUSUM and EWMA charts identified only two. This highlights the superior sensitivity of the CAM approach. In the second scenario ($\delta$ = 1.25, $\theta$ = 1.25), all three charts signaled the first OOC condition at sample 23. In this case, the CAM, CUSUM, and EWMA charts detected the same number of OOC points, which suggests that when shifts are large, the detection capabilities of all three methods converge. Additionally, no false alarms were observed in both cases.

**[Insert Figure 8 here]**

The OOC signals observed in the study can be linked to potential real-world operational issues in a CCPP. Variations in atmospheric pressure can be attributed to several underlying causes, including fluctuations in ambient weather conditions, irregularities in air intake systems, or inconsistencies in combustion efficiency. If a significant shift in AP variance is detected, it may indicate disruptions in the airflow of gas turbine, requiring immediate inspection of air filters, intake valves, or compressor performance. To complement the visual comparison in Figure 8, Table 3 summarizes the detection behavior of each chart under the two shift scenarios.

[Insert Table 3 Here]

The quality control team at the power plant should investigate any sudden shifts in AP to determine whether they result from environmental factors or operational inefficiencies. If the shifts coincide with external weather conditions, adjustments in combustion control settings may be necessary to stabilize power output. If internal system issues are suspected, maintenance on turbine components and airflow management systems should be prioritized. The ability of the CAM charts to detect these shifts earlier than existing methods provides a critical advantage, allowing plant operators to take preventive actions before process deviations escalate into severe performance losses. By integrating CAM-based monitoring into the control system of plant, real-time detection of AP fluctuations can be enhanced, leading to improved overall efficiency and reliability in power generation.

### 5.1 Practical and Managerial Implications

For practitioners, the proposed CAM charts provide a unified statistic for monitoring changes in both the process mean and variance, which eliminate the need to interpret separate charts for the two parameters. In a CCPP, a CAM signal should be viewed as a trigger for targeted investigation rather than as a direct diagnosis of a specific fault. Plant operators can compare the timing of the signal with operating conditions, such as atmospheric conditions, plant load,

and combustion behavior, to distinguish externally induced variations from possible equipment related disturbances. The CAM charts can therefore serve as an additional monitoring layer within existing plant monitoring systems, where the control limit is calibrated according to the desired IC performance. A previous study by [43] backs the integration of process monitoring information with maintenance decisions. It showed that SPC can support maintenance scheduling and improve the coordination between process control and maintenance activities.

From a managerial perspective, earlier identification of moderate process changes can provide additional time to prioritize inspections and maintenance resources before the deviations develop into more serious problems. Managers may also establish escalation procedures in which an initial CAM signal prompts verification of sensor readings and environmental conditions. This will be followed by equipment inspection when abnormal behavior persists. By using this procedure, the monitoring results can be used to support operational decisions instead of relying only on reactive interventions. Such use of continuous process information is consistent with recent Industry 4.0 maintenance literature [44], which highlights the role of data-driven monitoring in reducing downtime and operational costs while supporting productivity and decision making. These implications are not limited only to our case study. They are also relevant to other industrial processes in which simultaneous changes in process location and dispersion may affect efficiency of the systems.

## 6. Conclusion

This study introduced six memory-type control charts based on the CAM approach for the simultaneous monitoring of process mean and variance. These charts were developed using the maximum and sum of squares techniques and incorporated three different variance transformation methods, enhancing the flexibility and robustness of traditional control charting frameworks. This work focuses on integrating conditional aggregation mechanisms, improving sensitivity to shifts in process parameters, and extending control charting techniques to provide a unified monitoring structure for both location and dispersion. These enhancements make the proposed CAM-based charts a more effective approach for detecting and diagnosing process deviations compared to conventional methods. The performance evaluation conducted through Monte Carlo simulations provided strong empirical evidence of the superiority of the CAM charts, particularly in detecting moderate process shifts. The assessment of run-length characteristics, including ARL, CED, and EQL, confirmed that CAM-based monitoring enables earlier and more consistent detection of process variations while maintaining a low false alarm rate. Unlike traditional joint monitoring approaches such as CUSUM and EWMA, which often struggle with balancing sensitivity and stability, CAM charts demonstrated an improved ability to detect process deviations efficiently while reducing unnecessary alarms, making them well-suited for industrial applications requiring high precision and reliability. The practical utility of the proposed charts was validated through a real-world case study in a CCPP process, where CAM-based monitoring proved to be significantly more responsive in identifying shifts in atmospheric pressure, a critical variable affecting combustion efficiency and overall power generation performance. The ability of the CAM charts to detect subtle yet meaningful process shifts earlier than competing methods reinforces their potential for widespread adoption in industries where maintaining process stability, minimizing variability, and ensuring optimal performance are paramount.

Future research should explore ways to further enhance the applicability of CAM charts in diverse industrial settings by evaluating their performance under varying sample sizes, parameter estimation conditions, and process dynamics. Since this study assumes normality, future work could extend the CAM framework to non-parametric or distribution-free approaches, making it more applicable to real-world processes with non-Gaussian characteristics. Additionally, integrating CAM charts into real-time adaptive monitoring systems and automated decision-making frameworks could improve industrial process control, particularly in smart manufacturing and production environments. Exploring hybrid SPC methodologies that combine CAM-based monitoring with multivariate process control techniques would further expand its utility in high-dimensional process monitoring. Investigating its potential for integration into Industry 4.0 frameworks, such as IoT-enabled predictive maintenance and machine learning-driven fault detection systems, could provide new opportunities for real-time optimization and early intervention in industrial processes.

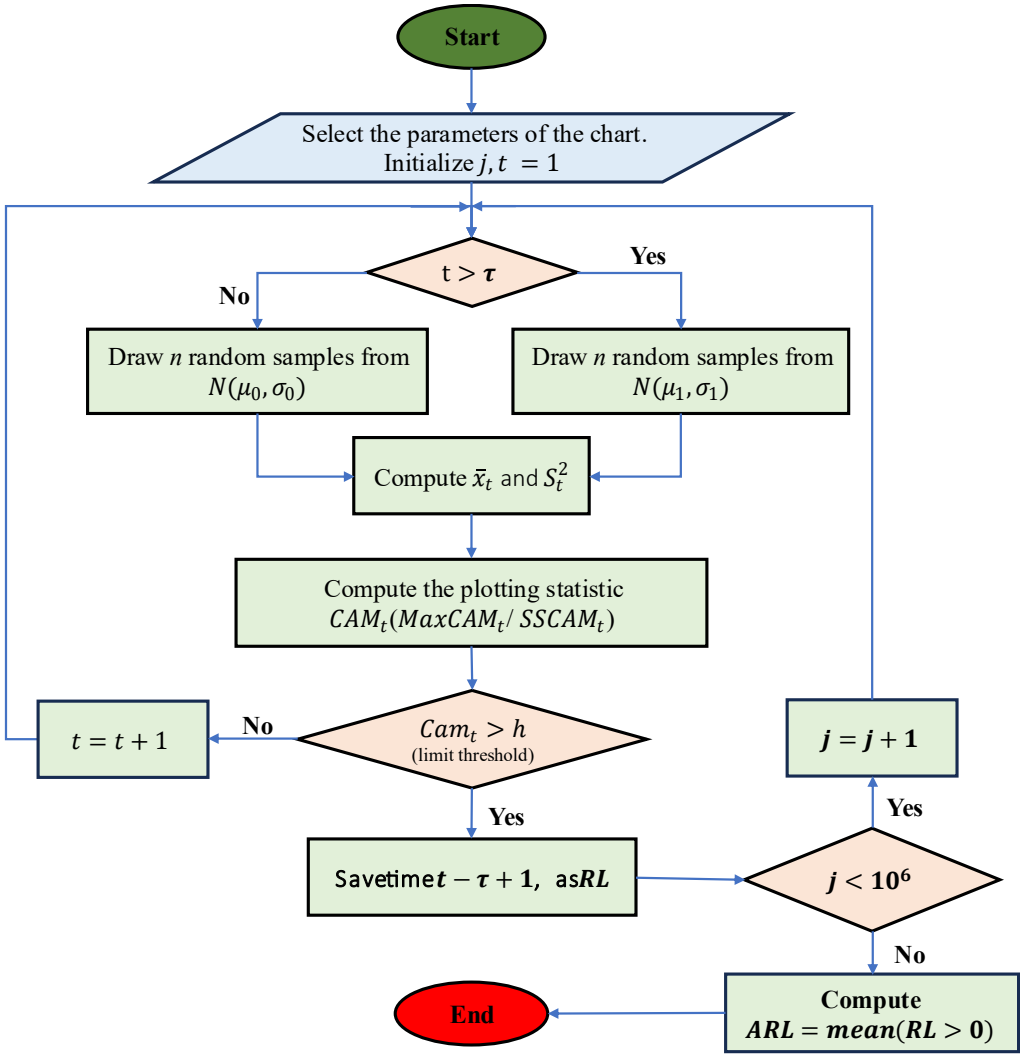
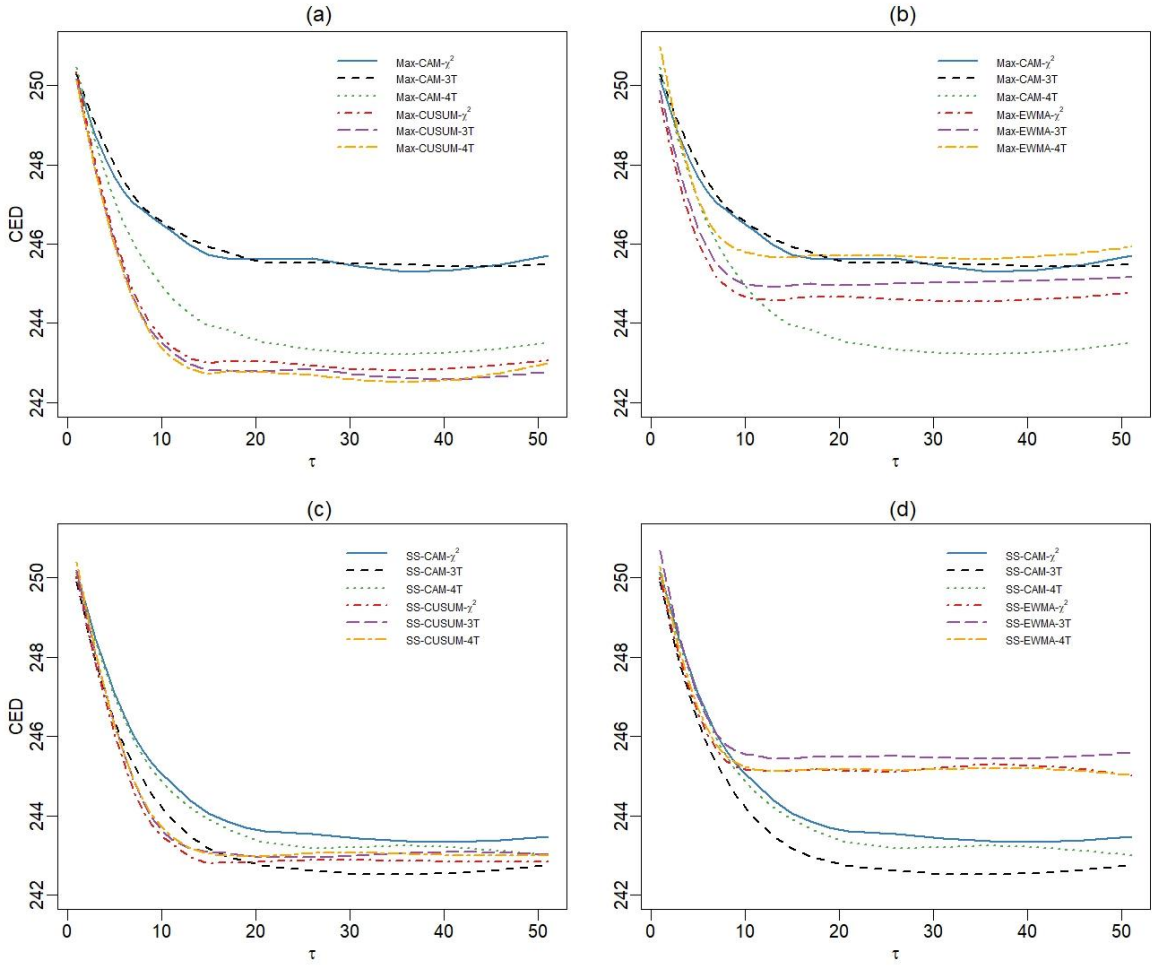



Figure 1: Flowchart for Computing ARL of the proposed charts




722
723 Figure 2: IC CED Curves of Proposed and Existing Charts

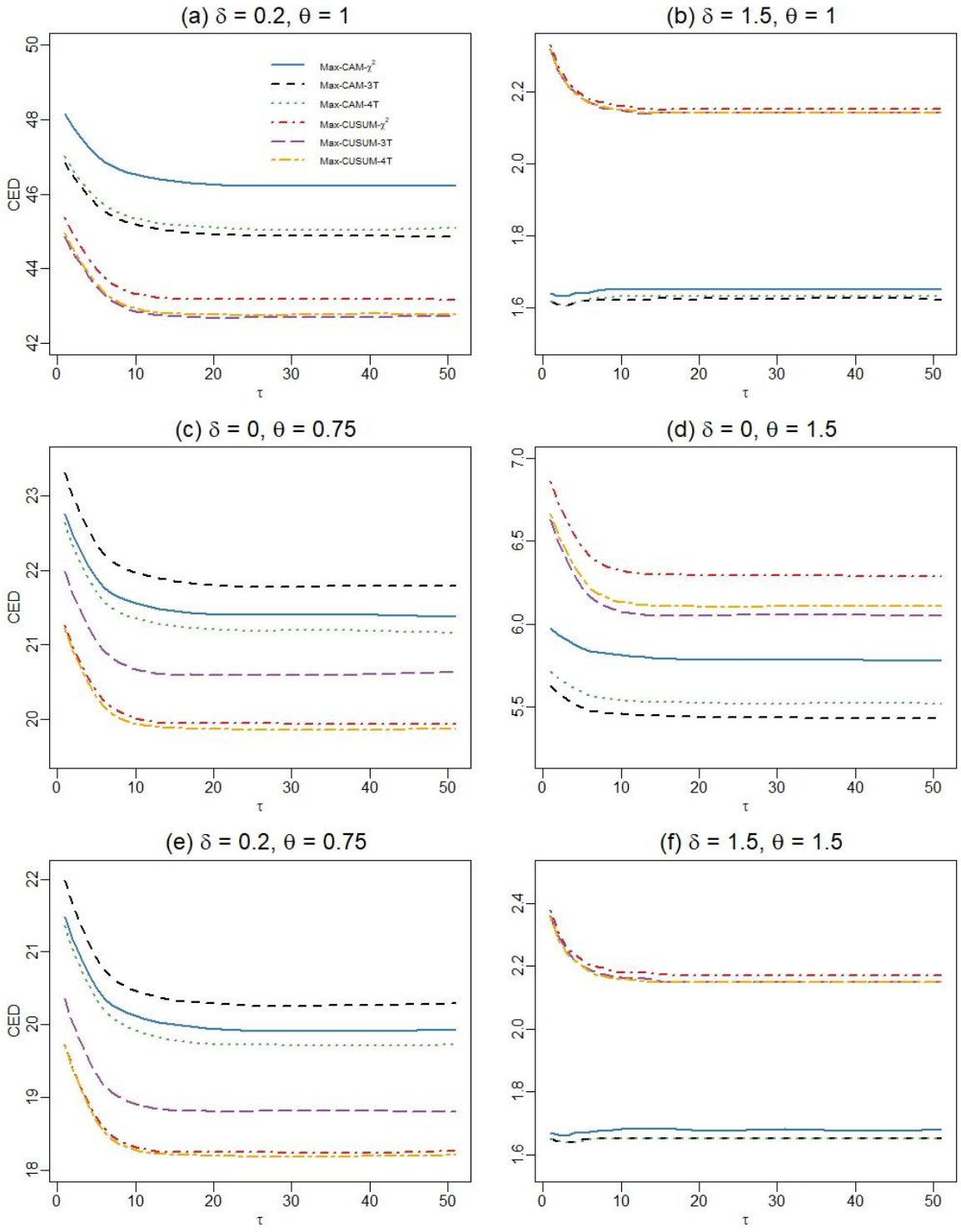



Figure 3: OOC CED Curves of Max-CAM and Max-CUSUM Charts.

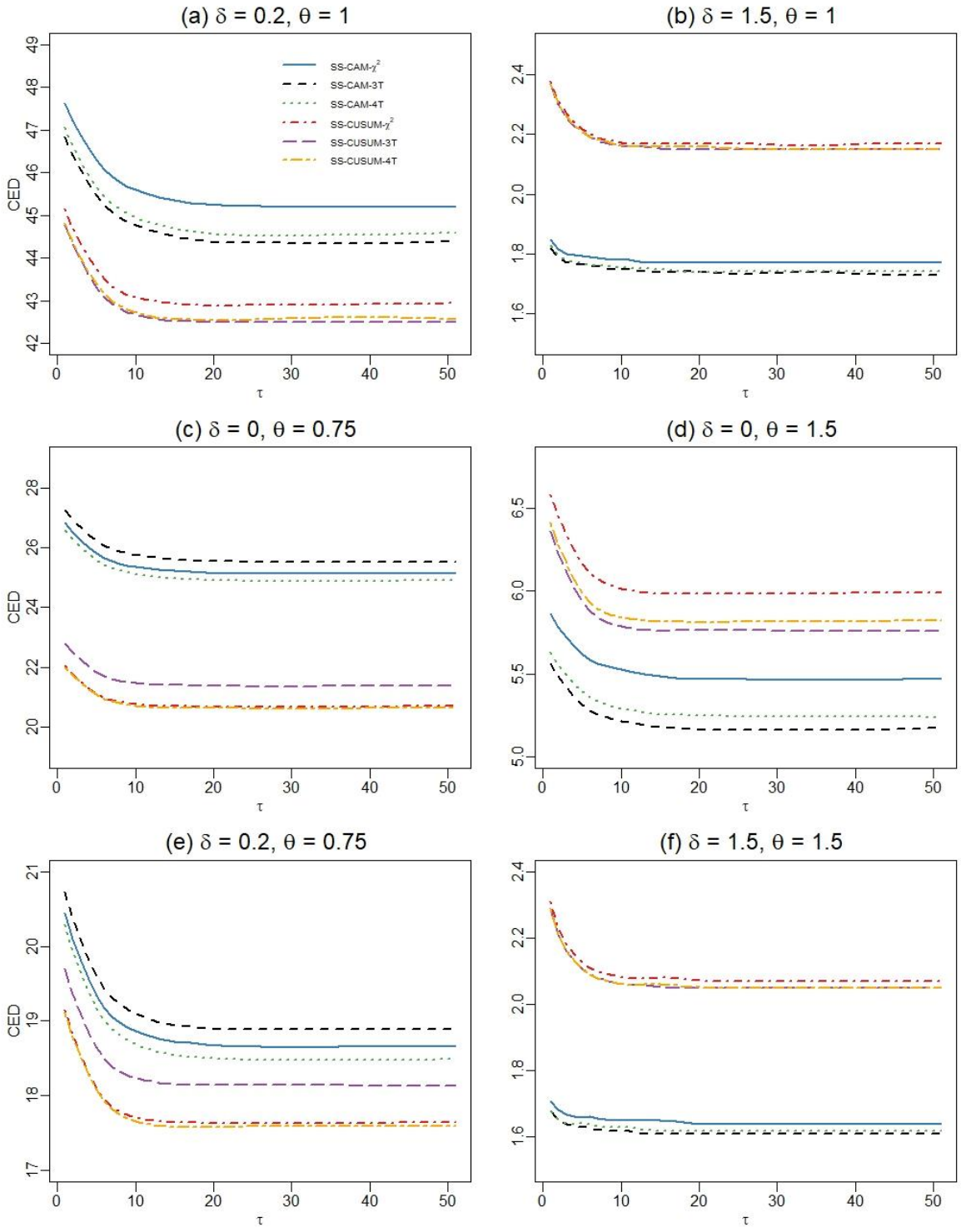



Figure 4: OOC CED Curves of SS-CAM and SS-CUSUM Charts.

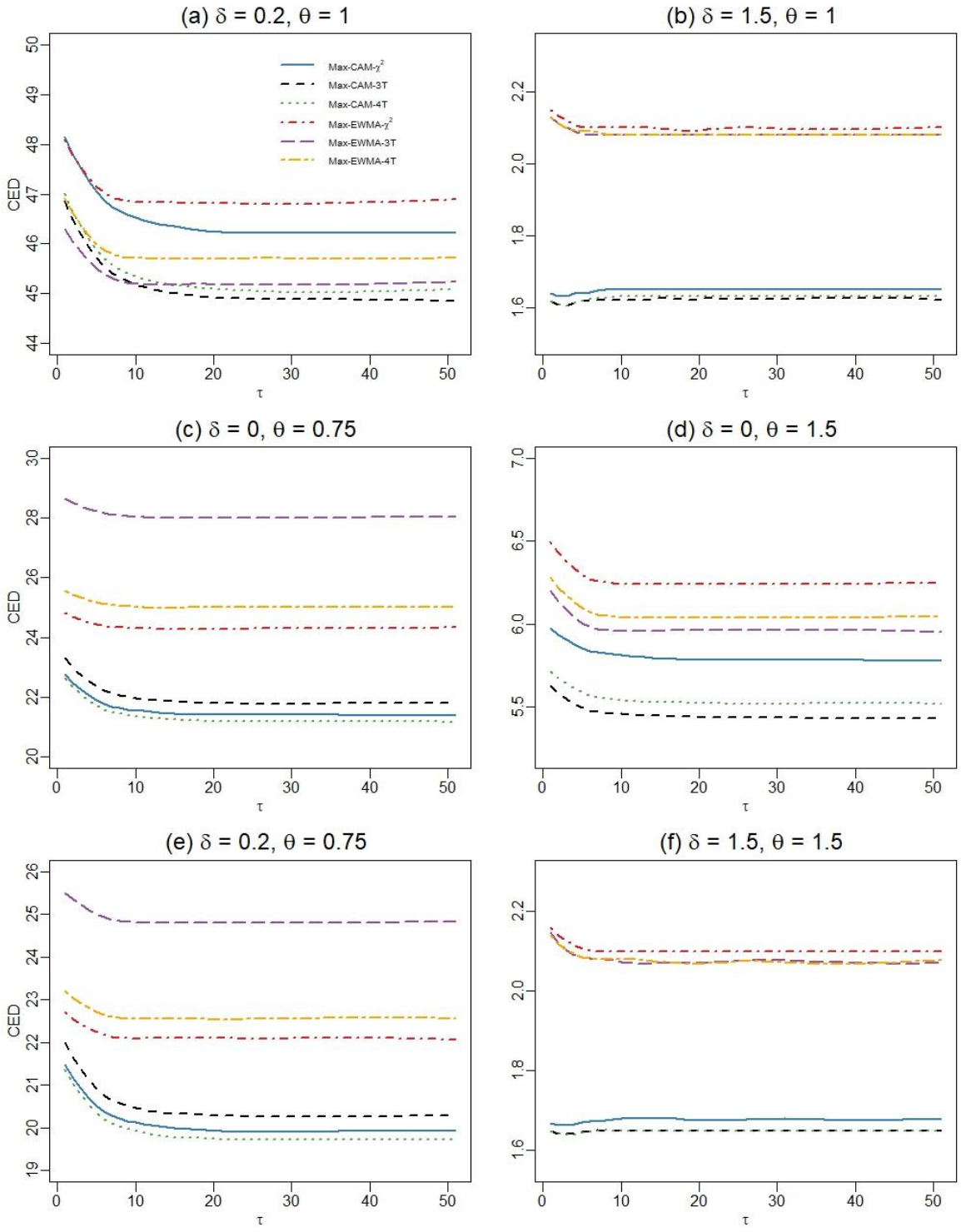



Figure 5: OOC CED Curves of Max-CAM and Max-EWMA Charts.

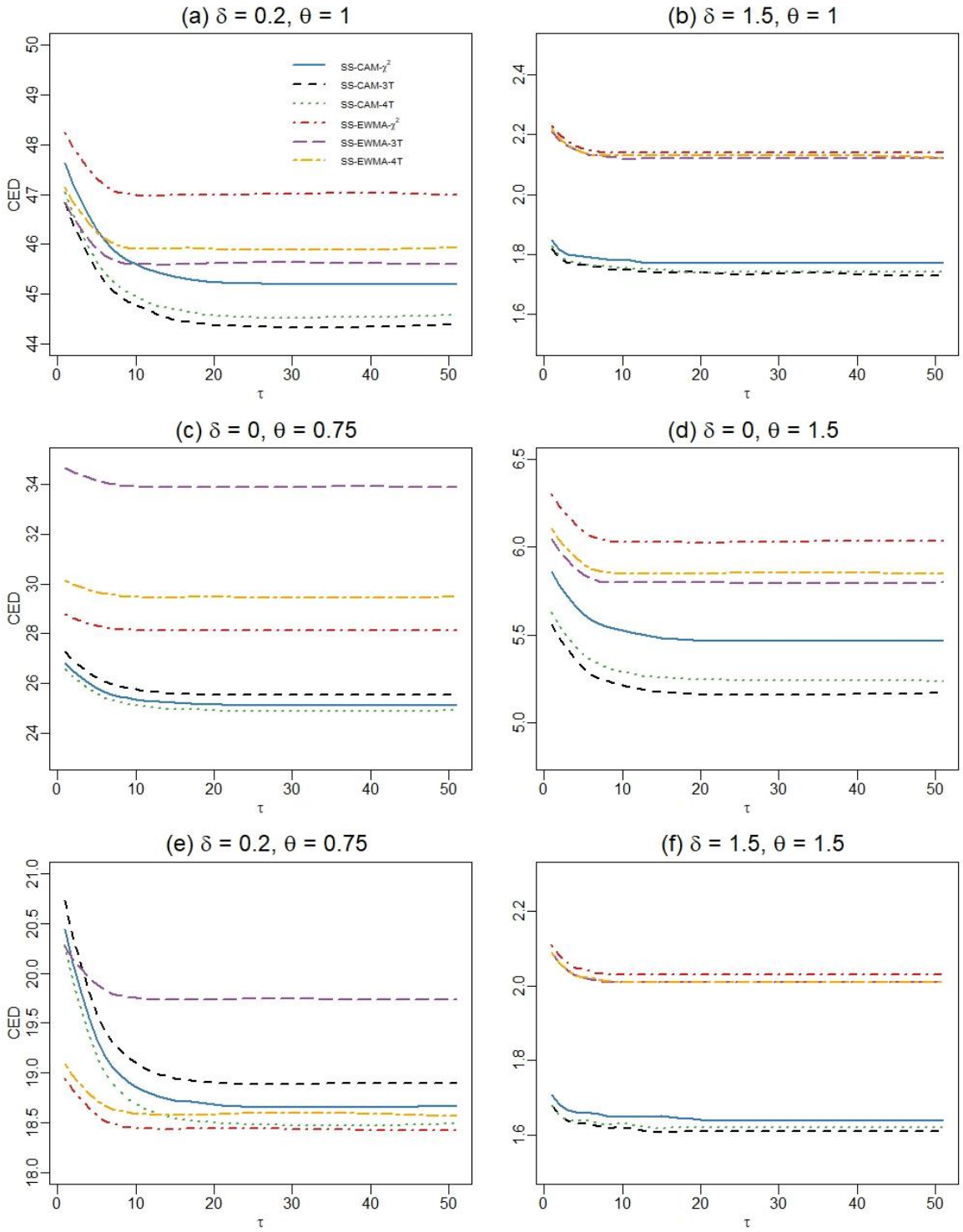



Figure 6: OOC CED Curves of SS-CAM and SS-EWMA Charts.

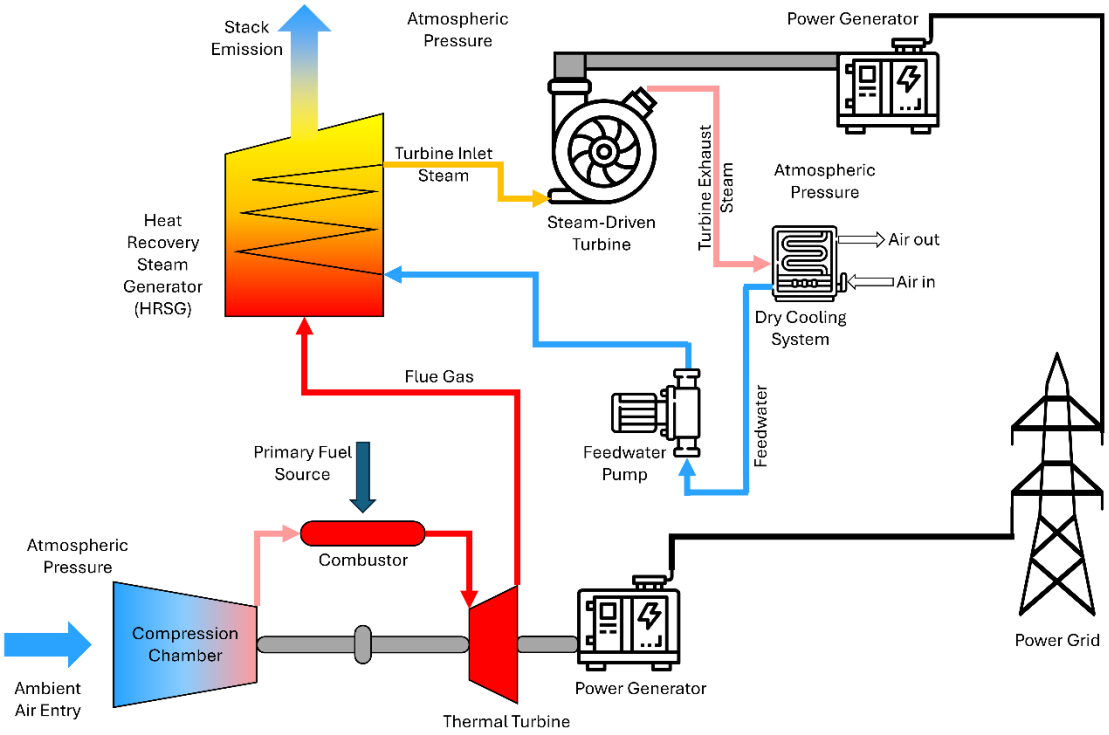



Figure 7: Atmospheric pressure influence in a Combined Cycle Power Plant

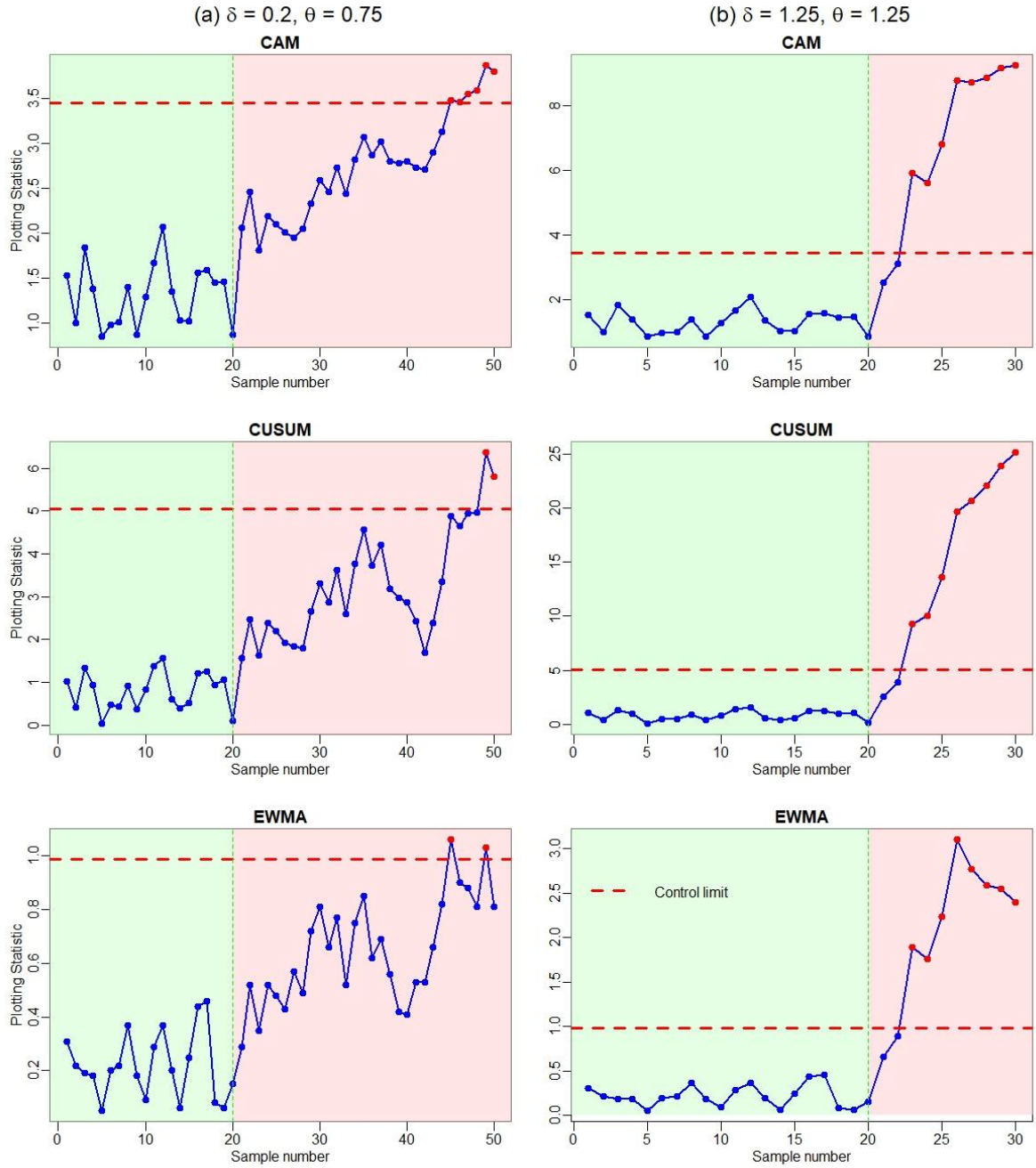



Figure 8: Application of Proposed and Existing Charts on AP Data.

Table 1. (ARL, SDRL, MDRL) values of the six proposed charts for different shifts in the process mean ($\delta$) and variance ($\theta$). $ARL_0 \cong 250$ and $k_2 = 0.5$.

| | | $\delta$ | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Chart | $\theta$ | 0.00 | 0.10 | 0.20 | 0.30 | 0.40 | 0.50 | 0.75 | 1.00 | 1.25 | 1.5 | 2.00 | 3.00 |
| Max-3T | 0.25 | (3.53,0.51,4) | (3.53,0.51,4) | (3.53,0.51,4) | (3.53,0.51,4) | (3.53,0.51,4) | (3.53,0.51,4) | (3.52,0.50,4) | (2.86,0.39,3) | (2.01,0.15,2) | (1.59,0.49,2) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.50 | (6.61,1.66,6) | (6.61,1.65,6) | (6.61,1.65,6) | (6.60,1.65,6) | (6.58,1.63,6) | (6.39,1.49,6) | (4.55,1.09,5) | (2.88,0.74,3) | (2.04,0.50,2) | (1.55,0.51,2) | (1.02,0.13,1) | (1.00,0.00,1) |
| | 0.75 | (23.44,11.87,21) | (23.37,11.78,21) | (22.12,10.55,20) | (17.92,7.47,17) | (13.01,5.13,12) | (9.33,3.65,9) | (4.71,1.79,5) | (2.91,1.06,3) | (2.06,0.72,2) | (1.57,0.57,2) | (1.08,0.27,1) | (1.00,0.00,1) |
| | 0.90 | (205.66,192.73,147) | (144.25,131.71,104) | (52.83,41.35,41) | (23.27,14.12,20) | (13.71,7.09,13) | (9.31,4.53,9) | (4.67,2.10,4) | (2.92,1.23,3) | (2.07,0.83,2) | (1.60,0.62,2) | (1.12,0.33,1) | (1.00,0.01,1) |
| | 1.00 | (250.60,245.84,175) | (135.21,128.38,96) | (46.97,38.76,36) | (21.81,14.62,19) | (13.11,7.60,12) | (9.01,4.87,8) | (4.62,2.27,4) | (2.92,1.34,3) | (2.09,0.90,2) | (1.62,0.66,2) | (1.15,0.36,1) | (1.00,0.02,1) |
| | 1.10 | (61.77,56.79,45) | (50.06,44.76,37) | (30.25,24.49,24) | (18.04,12.70,15) | (11.86,7.49,10) | (8.47,5.00,8) | (4.52,2.40,4) | (2.91,1.43,3) | (2.10,0.96,2) | (1.64,0.70,2) | (1.17,0.39,1) | (1.00,0.04,1) |
| | 1.25 | (15.95,12.11,13) | (15.19,11.45,12) | (13.33,9.74,11) | (10.91,7.61,9) | (8.70,5.78,7) | (6.94,4.40,6) | (4.21,2.43,4) | (2.84,1.51,3) | (2.10,1.03,2) | (1.66,0.75,2) | (1.21,0.42,1) | (1.00,0.06,1) |
| | 1.50 | (5.67,3.81,5) | (5.60,3.76,5) | (5.40,3.61,5) | (5.10,3.37,4) | (4.71,3.07,4) | (4.29,2.75,4) | (3.30,2.00,3) | (2.54,1.43,2) | (2.01,1.05,2) | (1.65,0.80,1) | (1.25,0.47,1) | (1.01,0.11,1) |
| | 1.75 | (3.27,2.06,3) | (3.25,2.04,3) | (3.20,2.01,3) | (3.12,1.94,3) | (3.01,1.86,3) | (2.87,1.76,2) | (2.51,1.48,2) | (2.14,1.20,2) | (1.83,0.96,2) | (1.59,0.77,1) | (1.26,0.50,1) | (1.03,0.16,1) |
| | 2.00 | (2.32,1.37,2) | (2.31,1.36,2) | (2.29,1.35,2) | (2.26,1.32,2) | (2.22,1.29,2) | (2.16,1.25,2) | (2.00,1.12,2) | (1.82,0.98,2) | (1.64,0.84,1) | (1.48,0.71,1) | (1.25,0.49,1) | (1.04,0.19,1) |
| Max-4T | 0.25 | (3.01,0.34,3) | (3.01,0.34,3) | (3.01,0.34,3) | (3.01,0.34,3) | (3.01,0.34,3) | (3.01,0.34,3) | (3.01,0.34,3) | (2.81,0.40,3) | (2.01,0.15,2) | (1.60,0.49,2) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.50 | (6.04,1.77,6) | (6.04,1.77,6) | (6.04,1.77,6) | (6.04,1.77,6) | (6.02,1.74,6) | (5.88,1.61,6) | (4.41,1.06,4) | (2.87,0.73,3) | (2.04,0.50,2) | (1.56,0.51,2) | (1.02,0.13,1) | (1.00,0.00,1) |
| | 0.75 | (22.77,12.14,20) | (22.70,12.07,20) | (21.51,10.82,20) | (17.49,7.62,17) | (12.80,5.15,12) | (9.25,3.63,9) | (4.71,1.78,5) | (2.92,1.06,3) | (2.06,0.72,2) | (1.57,0.57,2) | (1.08,0.27,1) | (1.00,0.00,1) |
| | 0.90 | (185.10,173.42,132) | (134.31,122.71,97) | (51.78,40.53,40) | (23.16,14.03,20) | (13.71,7.08,13) | (9.31,4.53,9) | (4.68,2.10,4) | (2.93,1.23,3) | (2.08,0.83,2) | (1.60,0.62,2) | (1.12,0.33,1) | (1.00,0.01,1) |
| | 1.00 | (249.61,245.03,174) | (135.55,128.80,96) | (47.18,38.99,36) | (21.88,14.67,19) | (13.15,7.63,12) | (9.03,4.88,8) | (4.63,2.28,4) | (2.93,1.34,3) | (2.09,0.90,2) | (1.62,0.66,2) | (1.15,0.36,1) | (1.00,0.02,1) |
| | 1.10 | (64.85,59.89,47) | (52.00,46.65,38) | (30.91,25.13,24) | (18.24,12.87,15) | (11.95,7.55,10) | (8.50,5.01,8) | (4.53,2.41,4) | (2.92,1.43,3) | (2.10,0.96,2) | (1.64,0.70,2) | (1.17,0.39,1) | (1.00,0.04,1) |
| | 1.25 | (16.41,12.50,13) | (15.64,11.79,13) | (13.65,9.98,11) | (11.11,7.76,9) | (8.82,5.89,8) | (7.01,4.47,6) | (4.22,2.45,4) | (2.85,1.52,3) | (2.10,1.03,2) | (1.66,0.75,2) | (1.21,0.43,1) | (1.00,0.07,1) |
| | 1.50 | (5.75,3.90,5) | (5.68,3.86,5) | (5.47,3.68,5) | (5.16,3.45,4) | (4.75,3.13,4) | (4.32,2.80,4) | (3.31,2.02,3) | (2.54,1.44,2) | (2.01,1.06,2) | (1.65,0.80,1) | (1.25,0.47,1) | (1.01,0.11,1) |
| | 1.75 | (3.28,2.10,3) | (3.26,2.09,3) | (3.20,2.05,3) | (3.12,1.99,3) | (3.01,1.90,3) | (2.88,1.79,2) | (2.50,1.50,2) | (2.13,1.21,2) | (1.82,0.97,2) | (1.58,0.77,1) | (1.26,0.49,1) | (1.03,0.16,1) |
| | 2.00 | (2.31,1.39,2) | (2.30,1.38,2) | (2.28,1.37,2) | (2.25,1.34,2) | (2.20,1.31,2) | (2.15,1.26,2) | (1.99,1.13,2) | (1.80,0.98,2) | (1.63,0.84,1) | (1.47,0.71,1) | (1.24,0.49,1) | (1.04,0.19,1) |
| Max-$\chi^2$ | 0.25 | (2.31,0.60,2) | (2.31,0.60,2) | (2.31,0.60,2) | (2.31,0.60,2) | (2.31,0.60,2) | (2.31,0.60,2) | (2.31,0.60,2) | (2.25,0.55,2) | (1.95,0.26,2) | (1.62,0.49,2) | (1.00,0.01,1) | (1.00,0.00,1) |
| | 0.50 | (5.75,2.03,6) | (5.75,2.03,6) | (5.74,2.03,6) | (5.74,2.03,6) | (5.73,2.00,6) | (5.60,1.86,5) | (4.29,1.16,4) | (2.87,0.74,3) | (2.06,0.51,2) | (1.58,0.50,2) | (1.02,0.14,1) | (1.00,0.00,1) |
| | 0.75 | (22.89,12.80,21) | (22.81,12.74,21) | (21.63,11.42,20) | (17.54,8.00,17) | (12.85,5.33,12) | (9.31,3.72,9) | (4.77,1.80,5) | (2.96,1.07,3) | (2.09,0.73,2) | (1.59,0.57,2) | (1.09,0.28,1) | (1.00,0.00,1) |
| | 0.90 | (170.25,160.43,121) | (128.72,118.56,93) | (51.84,40.80,40) | (23.40,14.20,21) | (13.90,7.18,13) | (9.46,4.58,9) | (4.76,2.12,4) | (2.97,1.24,3) | (2.11,0.84,2) | (1.62,0.63,2) | (1.13,0.34,1) | (1.00,0.01,1) |
| | 1.00 | (250.60,246.77,175) | (138.52,131.63,98) | (48.34,39.82,37) | (22.31,14.87,19) | (13.41,7.74,12) | (9.21,4.94,8) | (4.71,2.30,4) | (2.97,1.35,3) | (2.12,0.91,2) | (1.64,0.67,2) | (1.16,0.37,1) | (1.00,0.02,1) |
| | 1.10 | (70.81,65.44,51) | (56.15,50.44,41) | (32.54,26.43,25) | (18.89,13.28,16) | (12.29,7.71,11) | (8.71,5.10,8) | (4.62,2.45,4) | (2.96,1.45,3) | (2.13,0.97,2) | (1.66,0.71,2) | (1.18,0.40,1) | (1.00,0.04,1) |
| | 1.25 | (17.48,13.23,14) | (16.60,12.44,14) | (14.38,10.43,12) | (11.64,8.08,10) | (9.18,6.08,8) | (7.23,4.58,6) | (4.32,2.49,4) | (2.90,1.54,3) | (2.13,1.05,2) | (1.68,0.76,2) | (1.21,0.43,1) | (1.00,0.07,1) |
| | 1.50 | (6.01,4.05,5) | (5.93,3.99,5) | (5.72,3.83,5) | (5.37,3.56,5) | (4.94,3.24,4) | (4.49,2.89,4) | (3.41,2.08,3) | (2.60,1.48,2) | (2.05,1.08,2) | (1.68,0.81,1) | (1.26,0.48,1) | (1.01,0.12,1) |
| | 1.75 | (3.40,2.17,3) | (3.39,2.16,3) | (3.33,2.12,3) | (3.23,2.05,3) | (3.12,1.95,3) | (2.98,1.85,3) | (2.58,1.54,2) | (2.19,1.24,2) | (1.86,0.99,2) | (1.60,0.79,1) | (1.27,0.51,1) | (1.03,0.16,1) |
| | 2.00 | (2.38,1.43,2) | (2.37,1.42,2) | (2.35,1.41,2) | (2.32,1.38,2) | (2.27,1.34,2) | (2.21,1.30,2) | (2.04,1.16,2) | (1.85,1.01,2) | (1.66,0.86,1) | (1.50,0.72,1) | (1.26,0.50,1) | (1.04,0.20,1) |
| SS-3T | 0.25 | (4.42,0.50,4) | (4.39,0.50,4) | (4.26,0.45,4) | (4.07,0.33,4) | (3.85,0.40,4) | (3.47,0.50,3) | (2.90,0.31,3) | (2.08,0.27,2) | (1.99,0.12,2) | (1.52,0.50,2) | (1.00,0.01,1) | (1.00,0.00,1) |
| | 0.50 | (8.22,1.88,8) | (8.09,1.87,8) | (7.60,1.78,7) | (6.83,1.59,7) | (6.01,1.37,6) | (5.24,1.19,5) | (3.70,0.84,4) | (2.69,0.62,3) | (2.07,0.44,2) | (1.69,0.47,2) | (1.05,0.22,1) | (1.00,0.00,1) |
| | 0.75 | (27.38,12.95,25) | (25.80,12.40,24) | (20.87,9.99,19) | (15.16,6.55,14) | (11.17,4.37,11) | (8.52,3.15,8) | (4.87,1.68,5) | (3.18,1.05,3) | (2.30,0.73,2) | (1.78,0.58,2) | (1.17,0.38,1) | (1.00,0.01,1) |
| | 0.90 | (208.13,193.96,149) | (130.96,118.31,95) | (49.04,37.57,38) | (23.15,13.55,20) | (14.31,7.05,13) | (10.00,4.58,9) | (5.22,2.18,5) | (3.31,1.30,3) | (2.36,0.88,2) | (1.82,0.67,2) | (1.23,0.42,1) | (1.00,0.03,1) |
| | 1.00 | (249.99,243.50,175) | (131.41,123.19,94) | (46.90,37.59,36) | (22.66,14.56,20) | (14.04,7.75,13) | (9.82,5.03,9) | (5.16,2.39,5) | (3.30,1.42,3) | (2.36,0.96,2) | (1.82,0.71,2) | (1.25,0.44,1) | (1.00,0.04,1) |
| | 1.10 | (58.02,51.97,42) | (46.34,40.21,34) | (28.32,22.13,22) | (17.57,11.88,15) | (11.96,7.21,11) | (8.78,4.92,8) | (4.88,2.45,4) | (3.20,1.49,3) | (2.33,1.01,2) | (1.81,0.75,2) | (1.26,0.46,1) | (1.00,0.06,1) |
| | 1.25 | (15.25,11.07,13) | (14.40,10.34,12) | (12.39,8.63,10) | (10.13,6.71,9) | (8.18,5.14,7) | (6.66,3.99,6) | (4.24,2.31,4) | (2.96,1.49,3) | (2.23,1.05,2) | (1.77,0.78,2) | (1.28,0.48,1) | (1.01,0.09,1) |
| | 1.50 | (5.60,3.57,5) | (5.52,3.51,5) | (5.28,3.33,5) | (4.93,3.07,4) | (4.52,2.78,4) | (4.11,2.47,4) | (3.18,1.81,3) | (2.49,1.33,2) | (2.01,1.00,2) | (1.68,0.78,2) | (1.28,0.49,1) | (1.02,0.14,1) |
| | 1.75 | (3.28,1.97,3) | (3.26,1.95,3) | (3.20,1.90,3) | (3.10,1.84,3) | (2.97,1.74,3) | (2.83,1.64,2) | (2.44,1.37,2) | (2.09,1.11,2) | (1.79,0.90,2) | (1.56,0.74,1) | (1.26,0.49,1) | (1.03,0.17,1) |
| | 2.00 | (2.34,1.32,2) | (2.33,1.32,2) | (2.31,1.30,2) | (2.27,1.27,2) | (2.22,1.24,2) | (2.16,1.19,2) | (1.98,1.06,2) | (1.79,0.92,2) | (1.61,0.79,1) | (1.46,0.67,1) | (1.23,0.47,1) | (1.04,0.19,1) |
| SS-4T | 0.25 | (3.67,0.52,4) | (3.65,0.53,4) | (3.55,0.53,4) | (3.37,0.49,3) | (3.19,0.40,3) | (3.05,0.29,3) | (2.48,0.50,2) | (2.02,0.13,2) | (1.93,0.25,2) | (1.32,0.47,1) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.50 | (7.51,2.01,7) | (7.39,1.99,7) | (6.98,1.88,7) | (6.33,1.67,6) | (5.63,1.44,5) | (4.96,1.24,5) | (3.57,0.86,4) | (2.62,0.63,3) | (2.04,0.44,2) | (1.66,0.48,2) | (1.05,0.22,1) | (1.00,0.00,1) |
| | 0.75 | (26.72,13.26,24) | (25.18,12.67,23) | (20.42,10.14,18) | (14.90,6.65,14) | (11.03,4.43,10) | (8.43,3.20,8) | (4.84,1.70,5) | (3.17,1.06,3) | (2.30,0.74,2) | (1.78,0.58,2) | (1.18,0.38,1) | (1.00,0.01,1) |
| | 0.90 | (192.43,179.34,137) | (124.16,112.01,90) | (47.88,36.68,37) | (22.95,13.45,20) | (14.24,7.05,13) | (9.97,4.59,9) | (5.23,2.19,5) | (3.32,1.30,3) | (2.37,0.88,2) | (1.82,0.67,2) | (1.23,0.42,1) | (1.00,0.03,1) |
| | 1.00 | (250.37,243.81,175) | (131.65,123.32,94) | (47.14,37.79,36) | (22.75,14.60,20) | (14.08,7.75,13) | (9.86,5.04,9) | (5.18,2.39,5) | (3.31,1.42,3) | (2.37,0.96,2) | (1.83,0.72,2) | (1.25,0.44,1) | (1.00,0.04,1) |
| | 1.10 | (60.73,54.73,44) | (48.06,41.86,36) | (29.04,22.70,23) | (17.84,12.05,15) | (12.12,7.30,11) | (8.85,4.95,8) | (4.92,2.47,5) | (3.22,1.49,3) | (2.34,1.02,2) | (1.82,0.75,2) | (1.27,0.46,1) | (1.00,0.06,1) |
| | 1.25 | (15.67,11.38,13) | (14.79,10.67,12) | (12.69,8.86,11) | (10.33,6.87,9) | (8.32,5.24,7) | (6.75,4.05,6) | (4.28,2.33,4) | (2.99,1.50,3) | (2.24,1.05,2) | (1.78,0.79,2) | (1.28,0.48,1) | (1.01,0.09,1) |
| | 1.50 | (5.66,3.66,5) | (5.58,3.59,5) | (5.33,3.41,5) | (4.98,3.15,4) | (4.56,2.84,4) | (4.14,2.52,4) | (3.20,1.83,3) | (2.51,1.35,2) | (2.02,1.02,2) | (1.69,0.79,2) | (1.28,0.50,1) | (1.02,0.14,1) |
| | 1.75 | (3.28,2.01,3) | (3.25,2.00,3) | (3.19,1.95,3) | (3.10,1.87,3) | (2.97,1.78,3) | (2.82,1.67,2) | (2.44,1.39,2) | (2.09,1.13,2) | (1.80,0.91,2) | (1.57,0.74,1) | (1.26,0.49,1) | (1.03,0.17,1) |
| | 2.00 | (2.32,1.34,2) | (2.31,1.34,2) | (2.29,1.32,2) | (2.25,1.29,2) | (2.20,1.25,2) | (2.14,1.21,2) | (1.96,1.08,2) | (1.78,0.93,2) | (1.60,0.79,1) | (1.45,0.67,1) | (1.23,0.47,1) | (1.04,0.19,1) |
| SS-$\chi^2$ | 0.25 | (2.84,0.67,3) | (2.82,0.67,3) | (2.78,0.65,3) | (2.69,0.63,3) | (2.58,0.60,3) | (2.46,0.57,2) | (2.12,0.44,2) | (1.89,0.35,2) | (1.63,0.48,2) | (1.20,0.40,1) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.50 | (7.12,2.29,7) | (7.01,2.26,7) | (6.64,2.13,6) | (6.07,1.88,6) | (5.43,1.61,5) | (4.81,1.39,5) | (3.50,0.94,3) | (2.60,0.67,3) | (2.03,0.48,2) | (1.66,0.49,2) | (1.06,0.23,1) | (1.00,0.00,1) |
| | 0.75 | (26.94,13.95,25) | (25.38,13.31,23) | (20.55,10.59,19) | (15.01,6.89,14) | (11.10,4.58,11) | (8.49,3.30,8) | (4.89,1.74,5) | (3.21,1.08,3) | (2.32,0.75,2) | (1.80,0.59,2) | (1.19,0.39,1) | (1.00,0.01,1) |
| | 0.90 | (182.85,171.39,131) | (120.31,109.09,87) | (47.74,36.73,37) | (23.10,13.55,20) | (14.39,7.15,13) | (10.11,4.67,9) | (5.29,2.22,5) | (3.36,1.32,3) | (2.40,0.90,2) | (1.84,0.67,2) | (1.24,0.43,1) | (1.00,0.03,1) |
| | 1.00 | (249.96,243.54,175) | (133.03,124.81,95) | (47.81,38.30,37) | (23.09,14.77,20) | (14.31,7.85,13) | (10.02,5.11,9) | (5.26,2.43,5) | (3.36,1.44,3) | (2.40,0.97,2) | (1.85,0.72,2) | (1.26,0.45,1) | (1.00,0.05,1) |
| | 1.10 | (65.26,58.91,48) | (51.08,44.54,38) | (30.26,23.66,24) | (18.37,12.37,16) | (12.39,7.40,11) | (9.05,5.03,8) | (5.00,2.50,5) | (3.27,1.51,3) | (2.37,1.03,2) | (1.84,0.76,2) | (1.28,0.47,1) | (1.00,0.07,1) |
| | 1.25 | (16.49,11.92,14) | (15.56,11.14,13) | (13.26,9.18,11) | (10.73,7.08,9) | (8.58,5.36,7) | (6.94,4.13,6) | (4.37,2.37,4) | (3.04,1.52,3) | (2.28,1.07,2) | (1.81,0.80,2) | (1.29,0.49,1) | (1.01,0.10,1) |
| | 1.50 | (5.91,3.78,5) | (5.81,3.71,5) | (5.54,3.51,5) | (5.15,3.22,4) | (4.72,2.90,4) | (4.27,2.58,4) | (3.28,1.87,3) | (2.56,1.37,2) | (2.06,1.03,2) | (1.71,0.80,2) | (1.29,0.50,1) | (1.02,0.14,1) |
| | 1.75 | (3.39,2.06,3) | (3.37,2.05,3) | (3.30,2.00,3) | (3.19,1.92,3) | (3.06,1.83,3) | (2.91,1.71,3) | (2.50,1.42,2) | (2.13,1.15,2) | (1.82,0.93,2) | (1.59,0.76,1) | (1.27,0.50,1) | (1.03,0.17,1) |
| | 2.00 | (2.39,1.38,2) | (2.38,1.37,2) | (2.35,1.35,2) | (2.31,1.33,2) | (2.26,1.29,2) | (2.20,1.24,2) | (2.01,1.10,2) | (1.81,0.95,2) | (1.63,0.81,1) | (1.47,0.69,1) | (1.25,0.48,1) | (1.04,0.19,1) |

Table 2: Zero-state ARL Comparison of Proposed and Existing Charts.

| | | | Method | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | Max | | | SS | | |
| | | | Transformation | | | Transformation | | |
| Chart | $\delta$ | $\theta$ | $\chi^2$ | 3T | 4T | $\chi^2$ | 3T | 4T |
| CAM | 0.00 | 1.00 | (250.60,246.77,175) | (250.60,245.84,175) | (249.61,245.03,174) | (249.96,243.54,175) | (249.99,243.50,175) | (250.37,243.81,175) |
| | 0.20 | 1.00 | (48.34,39.82,37) | (46.97,38.76,36) | (47.18,38.99,36) | (47.81,38.30,37) | (46.90,37.59,36) | (47.14,37.79,36) |
| | 1.50 | 1.00 | (1.64,0.67,2) | (1.62,0.66,2) | (1.62,0.66,2) | (1.85,0.72,2) | (1.82,0.71,2) | (1.83,0.72,2) |
| | 0.00 | 0.75 | (22.89,12.80,21) | (23.44,11.87,21) | (22.77,12.14,20) | (26.94,13.95,25) | (27.38,12.95,25) | (26.72,13.26,24) |
| | 0.00 | 1.50 | (6.01,4.05,5) | (5.67,3.81,5) | (5.75,3.90,5) | (5.91,3.78,5) | (5.60,3.57,5) | (5.66,3.66,5) |
| | 0.20 | 0.75 | (21.63,11.42,20) | (22.12,10.55,20) | (21.51,10.82,20) | (20.55,10.59,19) | (20.87,9.99,19) | (20.42,10.14,18) |
| | 1.50 | 1.50 | (1.68,0.81,1) | (1.65,0.80,1) | (1.65,0.80,1) | (1.71,0.80,2) | (1.68,0.78,2) | (1.69,0.79,2) |
| | | EQL | 17.10 | 16.80 | 16.74 | 17.38 | 17.14 | 17.11 |
| CUSUM | 0.00 | 1.00 | (250.31, 243.71, 176) | (250.12, 243.41, 175) | (249.94, 242.97, 175) | (249.87, 242.96, 175) | (250.10, 243.69, 175) | (250.01, 243.00, 176) |
| | 0.20 | 1.00 | (45.46, 38.34, 34) | (44.97, 37.93, 34) | (45.06, 38.01, 34) | (45.24, 38.14, 34) | (44.80, 37.69, 33) | (44.87, 37.71, 34) |
| | 1.50 | 1.00 | (2.33, 0.54, 2) | (2.32, 0.54, 2) | (2.32, 0.54, 2) | (2.38, 0.56, 2) | (2.37, 0.55, 2) | (2.37, 0.55, 2) |
| | 0.00 | 0.75 | (21.34, 13.54, 18) | (22.09, 13.14, 19) | (21.34, 13.10, 18) | (22.15, 13.90, 19) | (22.91, 13.51, 19) | (22.12, 13.48, 19) |
| | 0.00 | 1.50 | (6.92, 3.64, 6) | (6.68, 3.49, 6) | (6.72, 3.58, 6) | (6.63, 3.48, 6) | (6.41, 3.35, 6) | (6.46, 3.44, 6) |
| | 0.20 | 0.75 | (19.83, 11.90, 17) | (20.50, 11.56, 18) | (19.83, 11.55, 17) | (19.27, 11.54, 16) | (19.81, 11.20, 17) | (19.23, 11.16, 16) |
| | 1.50 | 1.50 | (2.38, 0.76, 2) | (2.37, 0.75, 2) | (2.36, 0.76, 2) | (2.31, 0.71, 2) | (2.29, 0.71, 2) | (2.29, 0.71, 2) |
| | | EQL | 17.51 | 17.46 | 17.35 | 17.36 | 17.31 | 17.21 |
| EWMA | 0.00 | 1.00 | (249.78, 245.44, 174) | (250.14, 245.46, 175) | (250.67, 246.45, 175) | (250.00, 245.31, 175) | (250.5, 245.98, 175) | (250.28, 246.01, 175) |
| | 0.20 | 1.00 | (48.13, 42.51, 35) | (46.33, 40.78, 34) | (46.91, 41.32, 34) | (48.25, 42.70, 35) | (46.8, 41.29, 34) | (47.22, 41.59, 35) |
| | 1.50 | 1.00 | (2.15, 0.51, 2) | (2.13, 0.51, 2) | (2.13, 0.51, 2) | (2.23, 0.53, 2) | (2.22, 0.52, 2) | (2.22, 0.52, 2) |
| | 0.00 | 0.75 | (24.81, 18.78, 19) | (28.68, 21.44, 22) | (25.6, 19.02, 20) | (28.83, 22.39, 22) | (34.69, 26.96, 27) | (30.14, 23.12, 23) |
| | 0.00 | 1.50 | (6.52, 3.80, 6) | (6.21, 3.56, 5) | (6.3, 3.7, 5) | (6.32, 3.67, 5) | (6.06, 3.47, 5) | (6.12, 3.57, 5) |
| | 0.20 | 0.75 | (22.73, 16.55, 18) | (25.52, 18.26, 20) | (23.24, 16.59, 19) | (18.95, 12.94, 15) | (20.29, 13.6, 16) | (19.1, 12.8, 16) |
| | 1.50 | 1.50 | (2.17, 0.75, 2) | (2.14, 0.74, 2) | (2.14, 0.74, 2) | (2.11, 0.70, 2) | (2.09, 0.69, 2) | (2.1, 0.7, 2) |
| | | EQL | 18.19 | 18.37 | 17.99 | 18.13 | 18.44 | |

Table 3: Summary of Detection Results for the Charts Used in the CCPP Example

| Chart | Shift Scenarios | First OOC Sample | Detection Delay | Number of OOC Points | False Alarms |
|---|---|---|---|---|---|
| | Scenario I | | | | |
| CAM | | 45 | 25 | 6 | 0 |
| CUSUM | | 48 | 28 | 2 | 0 |
| s | | 45 | 25 | 2 | 0 |
| | Scenario II | | | | |
| CAM | | 23 | 3 | 8 | 0 |
| CUSUM | | 23 | 3 | 8 | 0 |
| EWMA | | 23 | 3 | 8 | 0 |

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## Appendix A: Mathematical Proofs

The distribution of $CAM_t$ statistic conditional on $CUSUM_{t-1}$, is normal.

Proof: The $CAM_t$ statistic for the negative CUSUM case is defined as:

$$CAM_t^- = \frac{1}{K_t^-} V_t + \left(1 - \frac{1}{K_t^-}\right) CAM_{t-1}^-.$$

where $D_t^-$ is the number of consecutive samples for which $CUSUM_{t-1}^- > 0$. Since $V_t$ represents one of the standardized transformations in equations (2), (4), and (6), then $V_t$ observations are independent and standard normal:

$$V_t \sim N(0,1).$$

The $CAM$ statistic is a linear combination of variables that are normally distributed. By the properties of linear combinations of normal variables, any such combination will also follow a normal distribution.

Thus, since $CAM_t^-$ is a weighted average of $V_t$, it follows a normal distribution. Therefore, given $CUSUM_{t-1}^-$, $CAM_t^-$ follows a normal distribution:

$$CAM_t^-|_{CUSUM_{t-1}^-} \sim N(\mu_{CAM^-}, \sigma_{CAM^-}^2).$$

By similar reasoning, the $CAM_t$ statistic for the positive CUSUM is also normal.

**Lemma A.1:** Conditional Mean of $CAM_t^-$ given $CUSUM_{t-1}^-$ is $0$.

Proof: We begin by representing the $CAM$ statistic as the average of the previous $D_t^-$ values of $V$ starting from when the last CUSUM was zero:

$$CAM_t^-|_{CUSUM_{t-1}^-} = \frac{1}{K_t^-} \sum_{m=t-D_t^-+1}^{t} V_m.$$

Next, we apply the expectation operator to both sides:

$$\mathbb{E}\left(CAM_t^-|_{CUSUM_{t-1}^-}\right) = \frac{1}{K_t^-} \sum_{m=t-D_t^-+1}^{t} \mathbb{E}(V_m).$$

Substituting $\mathbb{E}(V_t) = 0$, yields:

$$\mathbb{E}\left(CAM_t^-|_{CUSUM_{t-1}^-}\right) = \frac{1}{K_t^-} \sum_{m=t-D_{I(t)}^-+1}^{t} (0).$$

$$\mathbb{E}\left(CAM_t^-|_{CUSUM_{t-1}^-}\right) = 0.$$

**Lemma A.2:** For a given $CUSUM_{t-1}^-$, the conditional variance of $CAM_t^-$ is $\frac{1}{K_t^-}$.

Proof: By representing the CAM statistic as the average of the previous $D_t^-$ values of $V$ starting from when the last CUSUM was zero and applying the variance, we have:

$$\mathbb{V}\left(CAM_t^-|_{CUSUM_{t-1}^-}\right) = \frac{1}{(K_t^-)^2} \mathbb{V}\left(\sum_{m=t-D_t^-+1}^{t} V_m\right).$$

For independent random variables, the variance of their sum, $V_m$ is given by the sum of their variances. Thus:

$$\mathbb{V}\left(CAM_t^-|_{CUSUM_{t-1}^-}\right) = \frac{1}{\left(K_{(j)}^-\right)^2} \sum_{m=t-D_t^-+1}^{t} \mathbb{V}(V_m) = \frac{1}{(K_t^-)^2} \sum_{m=t-D_t^-+1}^{t} (1).$$

The summation from $t - D_t^- + 1$ to $t$ is performed $D_t^-$ times. Therefore,

$$\mathbb{V}\left(CAM_t^-|_{CUSUM_{t-1}^-}\right) = \frac{1}{(K_t^-)^2}[K_t^-(1)]$$

$$= \frac{1}{K_t^-}.$$

**Lemma A.3:** By following the same approach, the distribution of $CAM_t^+$ statistic conditional on the positive $CUSUM_{t-1}^+$, is given as:

$$CAM_t^+|_{CUSUM_{t-1}^+} \sim N\left(0, \frac{1}{K_t^+}\right).$$

## Appendix B: Supplementary Results

**[Insert Tables B1 – B4 here]**

Table B1. Limit coefficients of the proposed charts to achieve $ARL_0 \cong 250$

| $\boldsymbol{n = 5}$ | | | | |
|---|---|---|---|---|
| Max | $k_2$ | $\chi^2$ | 3T | 4T |
| | 0.25 | 3.386 | 3.354 | 3.358 |
| | 0.5 | 3.453 | 3.414 | 3.419 |
| | 0.75 | 3.446 | 3.4 | 3.406 |
| | 1 | 3.415 | 3.364 | 3.369 |
| | 1.25 | 3.375 | 3.318 | 3.322 |
| | 1.5 | 3.331 | 3.269 | 3.273 |
| SS | $k_2$ | $\chi^2$ | 3T | 4T |
| | 0.25 | 15.48 | 15.23 | 15.26 |
| | 0.5 | 15.53 | 15.21 | 15.27 |
| | 0.75 | 14.97 | 14.61 | 14.67 |
| | 1 | 14.32 | 13.92 | 13.98 |
| | 1.25 | 13.7 | 13.25 | 13.32 |
| | 1.5 | 13.14 | 12.64 | 12.72 |

Table B2. (ARL, SDRL, MDRL) values of the six proposed CAM charts for different shifts in the process mean ($\delta$) and variance ($\theta$). $ARL_0 \cong 250$ and $k_2 = 0.25$.

| | | $\delta$ | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Chart | $\theta$ | 0.00 | 0.10 | 0.20 | 0.30 | 0.40 | 0.50 | 0.75 | 1.00 | 1.25 | 1.5 | 2.00 | 3.00 |
| Max-3T | 0.25 | (3.38,0.49,3) | (3.38,0.49,3) | (3.38,0.49,3) | (3.38,0.49,3) | (3.38,0.49,3) | (3.38,0.49,3) | (3.36,0.48,3) | (2.79,0.43,3) | (2.00,0.15,2) | (1.50,0.50,1) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.50 | (6.41,1.64,6) | (6.41,1.64,6) | (6.41,1.64,6) | (6.41,1.63,6) | (6.38,1.60,6) | (6.19,1.47,6) | (4.40,1.07,4) | (2.80,0.73,3) | (1.99,0.50,2) | (1.50,0.51,2) | (1.01,0.11,1) | (1.00,0.00,1) |
| | 0.75 | (21.96,9.68,21) | (21.89,9.62,21) | (20.90,8.92,20) | (17.27,6.93,16) | (12.66,5.01,12) | (9.09,3.60,9) | (4.59,1.77,4) | (2.84,1.04,3) | (2.00,0.71,2) | (1.53,0.56,1) | (1.07,0.25,1) | (1.00,0.00,1) |
| | 0.90 | (112.50,83.53,91) | (86.20,59.01,73) | (41.40,23.53,38) | (21.89,11.67,20) | (13.42,6.86,12) | (9.12,4.50,8) | (4.56,2.08,4) | (2.85,1.22,3) | (2.02,0.82,2) | (1.56,0.61,1) | (1.11,0.31,1) | (1.00,0.01,1) |
| | 1.00 | (250.43,244.34,176) | (111.51,95.75,86) | (40.40,27.11,35) | (20.89,12.70,19) | (12.90,7.41,12) | (8.87,4.87,8) | (4.51,2.26,4) | (2.85,1.32,3) | (2.04,0.89,2) | (1.58,0.65,2) | (1.13,0.34,1) | (1.00,0.02,1) |
| | 1.10 | (60.04,51.71,46) | (48.52,39.98,38) | (29.39,21.69,24) | (17.86,12.09,15) | (11.76,7.46,10) | (8.36,5.02,7) | (4.42,2.40,4) | (2.84,1.41,3) | (2.05,0.94,2) | (1.60,0.69,1) | (1.16,0.37,1) | (1.00,0.03,1) |
| | 1.25 | (16.11,12.17,13) | (15.43,11.61,13) | (13.49,9.97,11) | (11.01,7.87,9) | (8.69,5.95,7) | (6.87,4.50,6) | (4.12,2.43,4) | (2.78,1.49,2) | (2.05,1.01,2) | (1.63,0.74,1) | (1.19,0.41,1) | (1.00,0.06,1) |
| | 1.50 | (5.61,3.91,5) | (5.55,3.86,5) | (5.34,3.69,4) | (5.03,3.44,4) | (4.65,3.13,4) | (4.22,2.79,4) | (3.23,2.00,3) | (2.48,1.42,2) | (1.97,1.04,2) | (1.62,0.78,1) | (1.23,0.46,1) | (1.01,0.11,1) |
| | 1.75 | (3.21,2.07,3) | (3.19,2.05,3) | (3.14,2.02,3) | (3.05,1.95,3) | (2.95,1.86,2) | (2.82,1.76,2) | (2.45,1.47,2) | (2.10,1.19,2) | (1.79,0.95,2) | (1.56,0.76,1) | (1.25,0.48,1) | (1.02,0.15,1) |
| | 2.00 | (2.27,1.36,2) | (2.26,1.35,2) | (2.24,1.33,2) | (2.21,1.31,2) | (2.17,1.27,2) | (2.12,1.24,2) | (1.96,1.10,2) | (1.78,0.96,2) | (1.61,0.82,1) | (1.46,0.69,1) | (1.23,0.48,1) | (1.03,0.18,1) |
| Max-4T | 0.25 | (2.93,0.39,3) | (2.93,0.39,3) | (2.93,0.39,3) | (2.93,0.39,3) | (2.93,0.39,3) | (2.93,0.39,3) | (2.93,0.38,3) | (2.70,0.46,3) | (2.00,0.14,2) | (1.51,0.50,2) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.50 | (5.86,1.75,6) | (5.87,1.75,6) | (5.86,1.75,6) | (5.86,1.75,6) | (5.84,1.72,6) | (5.70,1.59,6) | (4.26,1.04,4) | (2.79,0.72,3) | (1.99,0.50,2) | (1.51,0.51,2) | (1.01,0.11,1) | (1.00,0.00,1) |
| | 0.75 | (21.44,10.05,20) | (21.36,10.03,20) | (20.41,9.24,19) | (16.88,7.09,16) | (12.45,5.04,12) | (9.01,3.59,9) | (4.58,1.76,4) | (2.84,1.04,3) | (2.01,0.71,2) | (1.53,0.56,2) | (1.07,0.25,1) | (1.00,0.00,1) |
| | 0.90 | (110.46,83.70,90) | (84.97,59.30,72) | (41.05,23.58,37) | (21.81,11.68,20) | (13.40,6.85,12) | (9.12,4.49,8) | (4.57,2.08,4) | (2.85,1.22,3) | (2.03,0.82,2) | (1.56,0.61,2) | (1.11,0.31,1) | (1.00,0.01,1) |
| | 1.00 | (250.05,244.08,176) | (111.70,96.11,86) | (40.57,27.20,35) | (20.92,12.73,19) | (12.92,7.43,12) | (8.89,4.88,8) | (4.52,2.26,4) | (2.86,1.32,3) | (2.04,0.89,2) | (1.58,0.65,2) | (1.13,0.35,1) | (1.00,0.02,1) |
| | 1.10 | (62.24,53.38,48) | (49.84,40.81,40) | (29.86,21.93,25) | (18.05,12.20,15) | (11.84,7.51,10) | (8.40,5.05,7) | (4.44,2.40,4) | (2.84,1.42,3) | (2.05,0.95,2) | (1.60,0.69,1) | (1.16,0.37,1) | (1.00,0.03,1) |
| | 1.25 | (16.52,12.43,14) | (15.81,11.86,13) | (13.77,10.18,11) | (11.20,8.01,9) | (8.80,6.03,7) | (6.93,4.55,6) | (4.13,2.45,4) | (2.78,1.50,2) | (2.05,1.02,2) | (1.63,0.74,1) | (1.19,0.41,1) | (1.00,0.06,1) |
| | 1.50 | (5.69,3.99,5) | (5.61,3.93,5) | (5.41,3.77,5) | (5.09,3.52,4) | (4.70,3.19,4) | (4.26,2.84,4) | (3.24,2.02,3) | (2.48,1.43,2) | (1.97,1.04,2) | (1.62,0.78,1) | (1.23,0.46,1) | (1.01,0.11,1) |
| | 1.75 | (3.22,2.11,3) | (3.20,2.09,3) | (3.15,2.06,3) | (3.06,1.99,3) | (2.95,1.90,2) | (2.82,1.79,2) | (2.45,1.49,2) | (2.09,1.20,2) | (1.79,0.95,2) | (1.55,0.76,1) | (1.24,0.48,1) | (1.02,0.15,1) |
| | 2.00 | (2.27,1.38,2) | (2.26,1.37,2) | (2.24,1.35,2) | (2.20,1.33,2) | (2.16,1.29,2) | (2.11,1.25,2) | (1.95,1.12,2) | (1.77,0.97,2) | (1.60,0.82,1) | (1.45,0.69,1) | (1.23,0.47,1) | (1.03,0.18,1) |
| Max-$\chi^2$ | 0.25 | (2.24,0.59,2) | (2.24,0.59,2) | (2.24,0.59,2) | (2.24,0.59,2) | (2.24,0.59,2) | (2.24,0.59,2) | (2.24,0.59,2) | (2.17,0.53,2) | (1.93,0.28,2) | (1.51,0.50,2) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.50 | (5.57,2.00,5) | (5.57,2.00,5) | (5.57,2.00,5) | (5.57,2.00,5) | (5.55,1.97,5) | (5.42,1.83,5) | (4.13,1.14,4) | (2.78,0.73,3) | (2.00,0.50,2) | (1.53,0.51,2) | (1.01,0.12,1) | (1.00,0.00,1) |
| | 0.75 | (21.55,10.67,20) | (21.45,10.62,20) | (20.49,9.77,19) | (16.90,7.46,16) | (12.46,5.22,12) | (9.03,3.67,9) | (4.63,1.78,4) | (2.87,1.05,3) | (2.03,0.72,2) | (1.55,0.56,2) | (1.07,0.26,1) | (1.00,0.00,1) |
| | 0.90 | (110.31,85.58,89) | (85.07,60.81,72) | (41.28,24.05,37) | (22.00,11.84,20) | (13.54,6.93,13) | (9.23,4.54,9) | (4.62,2.10,4) | (2.88,1.23,3) | (2.05,0.83,2) | (1.58,0.62,2) | (1.11,0.32,1) | (1.00,0.01,1) |
| | 1.00 | (250.06,245.18,175) | (113.46,97.56,87) | (41.12,27.48,36) | (21.24,12.87,19) | (13.12,7.50,12) | (9.01,4.92,8) | (4.58,2.28,4) | (2.89,1.34,3) | (2.07,0.89,2) | (1.60,0.66,2) | (1.14,0.35,1) | (1.00,0.02,1) |
| | 1.10 | (66.06,56.30,51) | (52.53,42.74,42) | (30.97,22.55,26) | (18.53,12.43,16) | (12.12,7.63,11) | (8.57,5.12,8) | (4.50,2.43,4) | (2.88,1.43,3) | (2.08,0.95,2) | (1.62,0.69,2) | (1.16,0.38,1) | (1.00,0.03,1) |
| | 1.25 | (17.41,12.93,14) | (16.59,12.27,14) | (14.44,10.54,12) | (11.65,8.26,10) | (9.10,6.20,8) | (7.14,4.66,6) | (4.23,2.50,4) | (2.82,1.52,3) | (2.08,1.03,2) | (1.64,0.75,2) | (1.20,0.42,1) | (1.00,0.06,1) |
| | 1.50 | (5.93,4.12,5) | (5.85,4.05,5) | (5.63,3.89,5) | (5.28,3.62,4) | (4.86,3.28,4) | (4.40,2.92,4) | (3.32,2.07,3) | (2.53,1.46,2) | (2.00,1.06,2) | (1.64,0.80,1) | (1.24,0.47,1) | (1.01,0.11,1) |
| | 1.75 | (3.33,2.17,3) | (3.31,2.16,3) | (3.25,2.12,3) | (3.16,2.04,3) | (3.04,1.95,3) | (2.91,1.84,2) | (2.52,1.53,2) | (2.14,1.23,2) | (1.82,0.97,2) | (1.57,0.77,1) | (1.25,0.49,1) | (1.02,0.16,1) |
| | 2.00 | (2.33,1.41,2) | (2.32,1.41,2) | (2.30,1.39,2) | (2.26,1.36,2) | (2.22,1.33,2) | (2.16,1.28,2) | (1.99,1.15,2) | (1.81,0.99,2) | (1.63,0.84,1) | (1.47,0.71,1) | (1.24,0.48,1) | (1.04,0.19,1) |
| SS-3T | 0.25 | (4.42,0.50,4) | (4.36,0.49,4) | (4.22,0.42,4) | (4.07,0.32,4) | (3.86,0.40,4) | (3.47,0.50,3) | (2.90,0.31,3) | (2.08,0.27,2) | (1.99,0.12,2) | (1.53,0.50,2) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.50 | (8.19,1.87,8) | (8.01,1.83,8) | (7.51,1.71,7) | (6.81,1.55,7) | (6.03,1.37,6) | (5.26,1.19,5) | (3.71,0.84,4) | (2.69,0.62,3) | (2.07,0.44,2) | (1.69,0.47,2) | (1.05,0.22,1) | (1.00,0.00,1) |
| | 0.75 | (26.29,10.67,25) | (24.31,9.86,23) | (19.61,7.75,19) | (14.88,5.69,14) | (11.21,4.18,11) | (8.58,3.12,8) | (4.89,1.69,5) | (3.19,1.05,3) | (2.30,0.73,2) | (1.79,0.58,2) | (1.18,0.38,1) | (1.00,0.01,1) |
| | 0.90 | (112.59,80.14,93) | (77.60,51.41,66) | (38.56,20.78,35) | (22.16,10.95,21) | (14.36,6.74,13) | (10.10,4.55,9) | (5.25,2.19,5) | (3.32,1.30,3) | (2.37,0.88,2) | (1.82,0.67,2) | (1.23,0.42,1) | (1.00,0.03,1) |
| | 1.00 | (249.74,237.18,178) | (106.90,87.51,83) | (41.51,26.09,37) | (22.50,12.79,20) | (14.31,7.64,13) | (10.00,5.08,9) | (5.20,2.41,5) | (3.31,1.43,3) | (2.36,0.96,2) | (1.82,0.72,2) | (1.25,0.44,1) | (1.00,0.04,1) |
| | 1.10 | (57.33,46.18,46) | (45.32,34.65,37) | (28.21,19.29,24) | (18.04,11.33,16) | (12.35,7.29,11) | (9.00,5.04,8) | (4.93,2.49,5) | (3.22,1.50,3) | (2.33,1.02,2) | (1.81,0.75,2) | (1.26,0.46,1) | (1.00,0.06,1) |
| | 1.25 | (16.18,11.35,14) | (15.28,10.66,13) | (13.10,8.94,11) | (10.65,7.01,9) | (8.52,5.39,7) | (6.88,4.18,6) | (4.30,2.37,4) | (2.99,1.51,3) | (2.24,1.05,2) | (1.78,0.79,2) | (1.28,0.48,1) | (1.01,0.09,1) |
| | 1.50 | (5.80,3.80,5) | (5.72,3.73,5) | (5.46,3.54,5) | (5.08,3.26,4) | (4.65,2.93,4) | (4.20,2.59,4) | (3.22,1.87,3) | (2.51,1.36,2) | (2.02,1.02,2) | (1.68,0.79,2) | (1.28,0.49,1) | (1.02,0.14,1) |
| | 1.75 | (3.33,2.04,3) | (3.32,2.03,3) | (3.25,1.98,3) | (3.15,1.91,3) | (3.01,1.81,3) | (2.86,1.70,2) | (2.46,1.40,2) | (2.10,1.13,2) | (1.80,0.91,2) | (1.57,0.74,1) | (1.26,0.49,1) | (1.03,0.17,1) |
| | 2.00 | (2.36,1.36,2) | (2.35,1.35,2) | (2.33,1.33,2) | (2.29,1.30,2) | (2.23,1.26,2) | (2.17,1.22,2) | (1.99,1.08,2) | (1.79,0.93,2) | (1.61,0.80,1) | (1.46,0.67,1) | (1.23,0.47,1) | (1.04,0.19,1) |
| SS-4T | 0.25 | (3.67,0.53,4) | (3.63,0.53,4) | (3.52,0.52,4) | (3.36,0.49,3) | (3.19,0.40,3) | (3.05,0.29,3) | (2.47,0.50,2) | (2.02,0.13,2) | (1.93,0.25,2) | (1.32,0.47,1) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.50 | (7.47,1.99,7) | (7.32,1.95,7) | (6.91,1.82,7) | (6.32,1.64,6) | (5.65,1.44,6) | (4.98,1.25,5) | (3.57,0.87,4) | (2.62,0.63,3) | (2.04,0.44,2) | (1.66,0.48,2) | (1.05,0.22,1) | (1.00,0.00,1) |
| | 0.75 | (25.69,11.04,24) | (23.80,10.20,22) | (19.27,7.97,18) | (14.65,5.83,14) | (11.06,4.25,11) | (8.49,3.16,8) | (4.86,1.70,5) | (3.17,1.06,3) | (2.29,0.74,2) | (1.78,0.58,2) | (1.18,0.38,1) | (1.00,0.01,1) |
| | 0.90 | (111.49,80.85,91) | (76.71,51.57,65) | (38.26,20.92,35) | (22.04,11.00,20) | (14.30,6.76,13) | (10.06,4.56,9) | (5.24,2.19,5) | (3.32,1.30,3) | (2.36,0.88,2) | (1.82,0.67,2) | (1.23,0.42,1) | (1.00,0.03,1) |
| | 1.00 | (249.68,237.17,177) | (106.84,87.51,83) | (41.55,26.11,37) | (22.54,12.80,20) | (14.32,7.64,13) | (10.01,5.09,9) | (5.21,2.42,5) | (3.31,1.43,3) | (2.37,0.96,2) | (1.83,0.72,2) | (1.25,0.44,1) | (1.00,0.04,1) |

| Chart | $\theta$ | $\delta$ | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 0.00 | 0.10 | 0.20 | 0.30 | 0.40 | 0.50 | 0.75 | 1.00 | 1.25 | 1.5 | 2.00 | 3.00 |
| Max-3T | 0.25 | (3.38,0.49,3) | (3.38,0.49,3) | (3.38,0.49,3) | (3.38,0.49,3) | (3.38,0.49,3) | (3.38,0.49,3) | (3.36,0.48,3) | (2.79,0.43,3) | (2.00,0.15,2) | (1.50,0.50,1) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 1.10 | (58.95,47.36,47) | (46.34,35.28,38) | (28.65,19.50,24) | (18.22,11.41,16) | (12.45,7.32,11) | (9.06,5.06,8) | (4.96,2.50,5) | (3.23,1.51,3) | (2.34,1.02,2) | (1.82,0.75,2) | (1.27,0.46,1) | (1.00,0.06,1) |
| | 1.25 | (16.54,11.58,14) | (15.59,10.84,13) | (13.35,9.08,11) | (10.83,7.12,9) | (8.65,5.47,8) | (6.95,4.22,6) | (4.34,2.39,4) | (3.00,1.52,3) | (2.25,1.06,2) | (1.79,0.79,2) | (1.28,0.48,1) | (1.01,0.09,1) |
| | 1.50 | (5.85,3.88,5) | (5.76,3.81,5) | (5.50,3.60,5) | (5.12,3.32,4) | (4.68,2.98,4) | (4.23,2.63,4) | (3.24,1.89,3) | (2.52,1.37,2) | (2.02,1.02,2) | (1.69,0.79,2) | (1.28,0.50,1) | (1.02,0.14,1) |
| | 1.75 | (3.32,2.08,3) | (3.29,2.06,3) | (3.23,2.01,3) | (3.13,1.94,3) | (3.00,1.84,3) | (2.85,1.72,2) | (2.46,1.42,2) | (2.09,1.14,2) | (1.80,0.92,2) | (1.57,0.75,1) | (1.26,0.49,1) | (1.03,0.17,1) |
| | 2.00 | (2.33,1.37,2) | (2.32,1.36,2) | (2.30,1.35,2) | (2.26,1.32,2) | (2.21,1.28,2) | (2.15,1.23,2) | (1.97,1.09,2) | (1.78,0.94,2) | (1.61,0.80,1) | (1.45,0.67,1) | (1.23,0.47,1) | (1.04,0.19,1) |
| SS-$\chi^2$ | 0.25 | (2.83,0.67,3) | (2.81,0.66,3) | (2.76,0.65,3) | (2.68,0.62,3) | (2.57,0.60,3) | (2.45,0.57,2) | (2.12,0.44,2) | (1.88,0.35,2) | (1.62,0.49,2) | (1.19,0.39,1) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.50 | (7.07,2.27,7) | (6.94,2.22,7) | (6.57,2.07,6) | (6.04,1.85,6) | (5.43,1.62,5) | (4.81,1.39,5) | (3.49,0.95,3) | (2.59,0.67,3) | (2.02,0.48,2) | (1.65,0.49,2) | (1.05,0.23,1) | (1.00,0.00,1) |
| | 0.75 | (25.88,11.66,24) | (23.96,10.72,23) | (19.37,8.35,18) | (14.72,6.06,14) | (11.11,4.40,11) | (8.53,3.26,8) | (4.89,1.74,5) | (3.20,1.08,3) | (2.31,0.75,2) | (1.80,0.59,2) | (1.19,0.39,1) | (1.00,0.01,1) |
| | 0.90 | (112.63,82.95,92) | (77.21,52.64,65) | (38.46,21.24,35) | (22.22,11.20,21) | (14.43,6.87,13) | (10.17,4.64,10) | (5.30,2.23,5) | (3.36,1.32,3) | (2.39,0.89,2) | (1.84,0.67,2) | (1.24,0.43,1) | (1.00,0.03,1) |
| | 1.00 | (250.23,238.27,178) | (107.44,87.87,84) | (41.87,26.27,37) | (22.81,12.93,21) | (14.51,7.73,13) | (10.15,5.16,9) | (5.28,2.44,5) | (3.35,1.44,3) | (2.39,0.97,2) | (1.84,0.72,2) | (1.26,0.45,1) | (1.00,0.05,1) |
| | 1.10 | (61.91,49.43,50) | (48.20,36.51,40) | (29.48,19.86,25) | (18.65,11.57,16) | (12.70,7.42,11) | (9.22,5.12,8) | (5.04,2.53,5) | (3.28,1.52,3) | (2.37,1.03,2) | (1.84,0.76,2) | (1.28,0.47,1) | (1.00,0.07,1) |
| | 1.25 | (17.29,11.94,15) | (16.27,11.18,14) | (13.87,9.33,12) | (11.18,7.28,10) | (8.88,5.56,8) | (7.12,4.29,6) | (4.42,2.43,4) | (3.05,1.54,3) | (2.28,1.07,2) | (1.80,0.80,2) | (1.29,0.49,1) | (1.01,0.10,1) |
| | 1.50 | (6.08,3.98,5) | (5.98,3.91,5) | (5.71,3.71,5) | (5.30,3.40,5) | (4.83,3.06,4) | (4.35,2.69,4) | (3.31,1.93,3) | (2.56,1.39,2) | (2.05,1.04,2) | (1.71,0.80,2) | (1.29,0.50,1) | (1.02,0.14,1) |
| | 1.75 | (3.43,2.14,3) | (3.41,2.12,3) | (3.33,2.07,3) | (3.23,1.99,3) | (3.09,1.88,3) | (2.93,1.76,3) | (2.51,1.45,2) | (2.13,1.16,2) | (1.83,0.93,2) | (1.59,0.76,1) | (1.27,0.50,1) | (1.03,0.17,1) |
| | 2.00 | (2.40,1.41,2) | (2.39,1.40,2) | (2.37,1.38,2) | (2.32,1.35,2) | (2.27,1.31,2) | (2.21,1.26,2) | (2.01,1.12,2) | (1.81,0.96,2) | (1.63,0.82,1) | (1.47,0.69,1) | (1.24,0.48,1) | (1.04,0.20,1) |

Table B3. (ARL, SDRL, MDRL) values of the six proposed CAM charts for different shifts in the process mean $(\delta)$ and variance $(\theta)$. $ARL_0 \cong 250$ and $k_2 = 0.75$.

| Chart | $\theta$ | $\delta$ | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 0.00 | 0.10 | 0.20 | 0.30 | 0.40 | 0.50 | 0.75 | 1.00 | 1.25 | 1.5 | 2.00 | 3.00 |
| Max-3T | 0.25 | (3.49,0.51,3) | (3.49,0.51,3) | (3.49,0.51,3) | (3.49,0.51,3) | (3.49,0.51,3) | (3.49,0.51,3) | (3.48,0.50,3) | (2.84,0.40,3) | (2.01,0.15,2) | (1.57,0.49,2) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.50 | (6.55,1.66,6) | (6.56,1.66,6) | (6.56,1.66,6) | (6.56,1.66,6) | (6.52,1.63,6) | (6.33,1.49,6) | (4.51,1.08,4) | (2.86,0.74,3) | (2.02,0.50,2) | (1.54,0.51,2) | (1.02,0.13,1) | (1.00,0.00,1) |
| | 0.75 | (33.28,25.11,26) | (33.15,25.01,25) | (30.66,22.53,24) | (22.10,14.21,18) | (13.93,7.11,12) | (9.38,4.01,9) | (4.66,1.78,4) | (2.89,1.05,3) | (2.04,0.72,2) | (1.56,0.57,2) | (1.08,0.27,1) | (1.00,0.00,1) |
| | 0.90 | (355.36,347.78,249) | (237.68,230.90,167) | (82.52,76.00,59) | (29.90,23.87,23) | (14.92,9.68,13) | (9.42,5.09,9) | (4.62,2.08,4) | (2.90,1.22,3) | (2.06,0.83,2) | (1.59,0.62,2) | (1.12,0.32,1) | (1.00,0.01,1) |
| | 1.00 | (249.89,246.50,174) | (153.66,149.70,108) | (59.39,54.56,43) | (25.61,20.81,19) | (13.95,9.63,12) | (9.07,5.38,8) | (4.56,2.25,4) | (2.90,1.32,3) | (2.07,0.89,2) | (1.61,0.66,2) | (1.14,0.35,1) | (1.00,0.02,1) |
| | 1.10 | (64.32,61.08,46) | (53.16,49.75,38) | (33.16,29.40,24) | (19.39,15.60,15) | (12.20,8.67,10) | (8.46,5.35,7) | (4.46,2.37,4) | (2.88,1.41,3) | (2.09,0.95,2) | (1.63,0.70,2) | (1.17,0.38,1) | (1.00,0.04,1) |
| | 1.25 | (16.21,13.32,12) | (15.42,12.54,12) | (13.37,10.52,11) | (10.87,8.12,9) | (8.60,6.01,7) | (6.82,4.46,6) | (4.14,2.39,4) | (2.82,1.49,3) | (2.09,1.02,2) | (1.65,0.75,2) | (1.20,0.42,1) | (1.00,0.06,1) |
| | 1.50 | (5.54,3.75,5) | (5.46,3.69,5) | (5.28,3.53,4) | (4.97,3.28,4) | (4.60,2.99,4) | (4.20,2.67,4) | (3.25,1.94,3) | (2.51,1.41,2) | (2.00,1.04,2) | (1.65,0.79,1) | (1.24,0.47,1) | (1.01,0.11,1) |
| | 1.75 | (3.22,2.01,3) | (3.20,1.99,3) | (3.15,1.95,3) | (3.07,1.89,3) | (2.96,1.81,3) | (2.83,1.71,2) | (2.48,1.44,2) | (2.13,1.18,2) | (1.82,0.95,2) | (1.58,0.77,1) | (1.26,0.49,1) | (1.03,0.16,1) |
| | 2.00 | (2.30,1.34,2) | (2.29,1.33,2) | (2.27,1.32,2) | (2.24,1.29,2) | (2.20,1.26,2) | (2.14,1.22,2) | (1.98,1.10,2) | (1.81,0.96,2) | (1.63,0.83,1) | (1.48,0.70,1) | (1.24,0.49,1) | (1.04,0.19,1) |
| Max-4T | 0.25 | (2.99,0.35,3) | (2.99,0.35,3) | (2.99,0.35,3) | (2.99,0.35,3) | (2.99,0.35,3) | (2.99,0.35,3) | (2.99,0.34,3) | (2.79,0.41,3) | (2.01,0.14,2) | (1.58,0.49,2) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.50 | (5.99,1.77,6) | (5.99,1.77,6) | (5.99,1.77,6) | (5.99,1.77,6) | (5.97,1.74,6) | (5.82,1.60,6) | (4.37,1.05,4) | (2.86,0.73,3) | (2.03,0.50,2) | (1.54,0.51,2) | (1.02,0.13,1) | (1.00,0.00,1) |
| | 0.75 | (31.02,23.67,24) | (30.82,23.46,24) | (28.73,21.32,22) | (21.20,13.77,17) | (13.63,7.03,12) | (9.30,3.99,9) | (4.67,1.77,4) | (2.90,1.05,3) | (2.05,0.72,2) | (1.56,0.57,2) | (1.08,0.27,1) | (1.00,0.00,1) |
| | 0.90 | (285.63,278.91,200) | (205.36,198.98,144) | (78.89,72.47,57) | (29.61,23.58,23) | (14.90,9.67,13) | (9.42,5.08,9) | (4.63,2.08,4) | (2.91,1.22,3) | (2.07,0.83,2) | (1.59,0.62,2) | (1.12,0.33,1) | (1.00,0.01,1) |
| | 1.00 | (250.56,247.63,174) | (154.48,150.20,109) | (59.85,55.07,43) | (25.70,20.88,20) | (14.00,9.66,12) | (9.10,5.39,8) | (4.58,2.26,4) | (2.90,1.32,3) | (2.08,0.89,2) | (1.61,0.66,2) | (1.15,0.36,1) | (1.00,0.02,1) |
| | 1.10 | (68.12,64.91,48) | (55.75,52.37,40) | (34.17,30.40,25) | (19.70,15.90,15) | (12.31,8.77,10) | (8.50,5.37,7) | (4.47,2.38,4) | (2.89,1.41,3) | (2.09,0.96,2) | (1.63,0.70,2) | (1.17,0.39,1) | (1.00,0.04,1) |
| | 1.25 | (16.82,13.92,13) | (15.95,13.06,12) | (13.80,10.93,11) | (11.11,8.33,9) | (8.73,6.13,7) | (6.89,4.52,6) | (4.15,2.40,4) | (2.82,1.49,3) | (2.09,1.02,2) | (1.65,0.75,2) | (1.20,0.42,1) | (1.00,0.06,1) |
| | 1.50 | (5.63,3.86,5) | (5.56,3.80,5) | (5.36,3.63,5) | (5.04,3.37,4) | (4.65,3.06,4) | (4.24,2.73,4) | (3.26,1.97,3) | (2.51,1.42,2) | (2.00,1.05,2) | (1.64,0.79,1) | (1.24,0.47,1) | (1.01,0.11,1) |
| | 1.75 | (3.23,2.05,3) | (3.22,2.04,3) | (3.17,2.00,3) | (3.08,1.93,3) | (2.97,1.85,3) | (2.84,1.75,2) | (2.47,1.47,2) | (2.12,1.19,2) | (1.81,0.96,2) | (1.57,0.77,1) | (1.25,0.49,1) | (1.03,0.16,1) |
| | 2.00 | (2.29,1.36,2) | (2.28,1.36,2) | (2.26,1.34,2) | (2.23,1.32,2) | (2.19,1.28,2) | (2.13,1.24,2) | (1.97,1.12,2) | (1.79,0.97,2) | (1.62,0.83,1) | (1.47,0.70,1) | (1.24,0.48,1) | (1.04,0.19,1) |
| Max-$\chi^2$ | 0.25 | (2.30,0.60,2) | (2.30,0.60,2) | (2.30,0.60,2) | (2.30,0.60,2) | (2.30,0.60,2) | (2.30,0.60,2) | (2.30,0.60,2) | (2.24,0.55,2) | (1.95,0.26,2) | (1.61,0.49,2) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.50 | (5.71,2.03,6) | (5.71,2.03,6) | (5.71,2.03,6) | (5.71,2.03,6) | (5.69,2.00,6) | (5.56,1.86,5) | (4.27,1.15,4) | (2.86,0.74,3) | (2.05,0.50,2) | (1.57,0.50,2) | (1.02,0.14,1) | (1.00,0.00,1) |
| | 0.75 | (30.58,23.87,23) | (30.45,23.79,23) | (28.49,21.67,22) | (21.20,14.12,17) | (13.69,7.19,12) | (9.37,4.07,9) | (4.74,1.79,5) | (2.95,1.06,3) | (2.08,0.73,2) | (1.59,0.57,2) | (1.09,0.28,1) | (1.00,0.00,1) |
| | 0.90 | (232.08,226.91,163) | (180.42,174.94,127) | (77.59,71.46,56) | (29.98,23.90,23) | (15.13,9.75,13) | (9.59,5.14,9) | (4.72,2.11,4) | (2.96,1.24,3) | (2.10,0.84,2) | (1.61,0.63,2) | (1.13,0.34,1) | (1.00,0.01,1) |
| | 1.00 | (250.39,247.14,175) | (158.64,154.79,111) | (61.94,56.96,45) | (26.51,21.51,20) | (14.33,9.82,12) | (9.30,5.46,8) | (4.67,2.29,4) | (2.96,1.34,3) | (2.11,0.90,2) | (1.63,0.67,2) | (1.15,0.37,1) | (1.00,0.02,1) |
| | 1.10 | (76.14,72.81,54) | (61.49,57.88,44) | (36.64,32.61,27) | (20.67,16.69,16) | (12.76,9.06,11) | (8.74,5.50,8) | (4.57,2.42,4) | (2.94,1.43,3) | (2.12,0.97,2) | (1.65,0.70,2) | (1.18,0.39,1) | (1.00,0.04,1) |
| | 1.25 | (18.16,15.04,14) | (17.19,14.09,13) | (14.72,11.67,12) | (11.74,8.81,10) | (9.13,6.39,8) | (7.16,4.68,6) | (4.27,2.46,4) | (2.88,1.52,3) | (2.13,1.04,2) | (1.68,0.76,2) | (1.21,0.43,1) | (1.00,0.07,1) |
| | 1.50 | (5.92,4.03,5) | (5.84,3.96,5) | (5.62,3.78,5) | (5.29,3.51,5) | (4.86,3.18,4) | (4.41,2.82,4) | (3.37,2.03,3) | (2.58,1.45,2) | (2.04,1.07,2) | (1.67,0.81,1) | (1.25,0.48,1) | (1.01,0.12,1) |
| | 1.75 | (3.36,2.13,3) | (3.34,2.11,3) | (3.29,2.07,3) | (3.20,2.00,3) | (3.08,1.91,3) | (2.94,1.80,3) | (2.55,1.51,2) | (2.17,1.23,2) | (1.85,0.98,2) | (1.60,0.79,1) | (1.27,0.50,1) | (1.03,0.16,1) |
| | 2.00 | (2.36,1.41,2) | (2.35,1.40,2) | (2.34,1.39,2) | (2.30,1.36,2) | (2.25,1.32,2) | (2.20,1.28,2) | (2.03,1.15,2) | (1.84,1.00,2) | (1.65,0.85,1) | (1.49,0.72,1) | (1.25,0.50,1) | (1.04,0.19,1) |
| SS-3T | 0.25 | (4.25,0.44,4) | (4.24,0.43,4) | (4.19,0.41,4) | (4.05,0.39,4) | (3.72,0.48,4) | (3.31,0.47,3) | (2.81,0.40,3) | (2.04,0.19,2) | (1.97,0.18,2) | (1.39,0.49,1) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.50 | (7.96,1.86,8) | (7.88,1.86,8) | (7.56,1.85,7) | (6.86,1.76,7) | (5.93,1.50,6) | (5.08,1.22,5) | (3.57,0.82,4) | (2.60,0.61,3) | (2.02,0.43,2) | (1.63,0.49,2) | (1.04,0.19,1) | (1.00,0.00,1) |
| | 0.75 | (39.56,29.75,30) | (37.27,27.84,29) | (29.71,21.67,23) | (19.25,12.54,16) | (12.21,6.42,11) | (8.62,3.73,8) | (4.74,1.70,5) | (3.09,1.04,3) | (2.23,0.72,2) | (1.73,0.58,2) | (1.15,0.36,1) | (1.00,0.01,1) |
| | 0.90 | (367.57,359.96,258) | (226.59,219.62,159) | (78.83,72.10,57) | (29.90,23.47,23) | (15.42,9.71,13) | (9.96,5.15,9) | (5.05,2.16,5) | (3.21,1.27,3) | (2.29,0.87,2) | (1.77,0.66,2) | (1.20,0.40,1) | (1.00,0.02,1) |
| | 1.00 | (250.03,246.03,175) | (152.84,148.31,107) | (59.71,54.36,43) | (26.30,20.97,20) | (14.62,9.76,12) | (9.66,5.50,9) | (4.97,2.35,5) | (3.19,1.39,3) | (2.29,0.94,2) | (1.77,0.70,2) | (1.22,0.43,1) | (1.00,0.04,1) |
| | 1.10 | (60.63,56.85,43) | (49.62,45.79,36) | (31.10,27.10,23) | (18.62,14.62,14) | (12.06,8.33,10) | (8.57,5.24,7) | (4.70,2.41,4) | (3.09,1.45,3) | (2.26,0.99,2) | (1.76,0.74,2) | (1.24,0.44,1) | (1.00,0.06,1) |
| | 1.25 | (15.16,12.10,12) | (14.34,11.36,11) | (12.31,9.43,10) | (9.98,7.24,8) | (7.98,5.42,7) | (6.42,4.07,6) | (4.08,2.26,4) | (2.87,1.45,3) | (2.17,1.02,2) | (1.73,0.77,2) | (1.25,0.46,1) | (1.01,0.08,1) |
| | 1.50 | (5.33,3.49,5) | (5.25,3.41,4) | (5.03,3.25,4) | (4.71,2.99,4) | (4.33,2.70,4) | (3.94,2.39,3) | (3.06,1.74,3) | (2.41,1.29,2) | (1.96,0.98,2) | (1.64,0.76,1) | (1.26,0.48,1) | (1.02,0.13,1) |
| | 1.75 | (3.15,1.89,3) | (3.13,1.87,3) | (3.07,1.83,3) | (2.97,1.76,3) | (2.86,1.68,2) | (2.73,1.58,2) | (2.36,1.31,2) | (2.03,1.08,2) | (1.75,0.88,2) | (1.53,0.72,1) | (1.24,0.47,1) | (1.03,0.16,1) |
| | 2.00 | (2.27,1.27,2) | (2.26,1.27,2) | (2.24,1.25,2) | (2.20,1.23,2) | (2.15,1.19,2) | (2.10,1.15,2) | (1.92,1.03,2) | (1.75,0.90,2) | (1.58,0.77,1) | (1.43,0.65,1) | (1.22,0.46,1) | (1.03,0.18,1) |
| SS-4T | 0.25 | (3.53,0.53,4) | (3.51,0.53,3) | (3.46,0.52,3) | (3.32,0.48,3) | (3.14,0.37,3) | (2.99,0.31,3) | (2.35,0.48,2) | (2.01,0.10,2) | (1.87,0.33,2) | (1.22,0.41,1) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.50 | (7.26,1.97,7) | (7.19,1.97,7) | (6.91,1.94,7) | (6.33,1.81,6) | (5.54,1.53,5) | (4.81,1.26,5) | (3.44,0.85,3) | (2.54,0.62,2) | (1.99,0.45,2) | (1.60,0.50,2) | (1.04,0.18,1) | (1.00,0.00,1) |

| Chart | $\theta$ | $\delta$ | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 0.00 | 0.10 | 0.20 | 0.30 | 0.40 | 0.50 | 0.75 | 1.00 | 1.25 | 1.5 | 2.00 | 3.00 |
| Max-3T | 0.25 | (3.49,0.51,3) | (3.49,0.51,3) | (3.49,0.51,3) | (3.49,0.51,3) | (3.49,0.51,3) | (3.49,0.51,3) | (3.48,0.50,3) | (2.84,0.40,3) | (2.01,0.15,2) | (1.57,0.49,2) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.75 | (36.98,27.94,28) | (34.87,26.27,27) | (28.01,20.53,22) | (18.51,12.11,15) | (11.95,6.36,11) | (8.50,3.73,8) | (4.70,1.72,4) | (3.08,1.05,3) | (2.23,0.73,2) | (1.73,0.58,2) | (1.15,0.36,1) | (1.00,0.01,1) |
| | 0.90 | (304.60,297.79,213) | (198.69,192.02,140) | (74.44,67.77,54) | (29.26,22.92,22) | (15.30,9.63,13) | (9.93,5.16,9) | (5.05,2.17,5) | (3.21,1.28,3) | (2.29,0.87,2) | (1.77,0.66,2) | (1.20,0.41,1) | (1.00,0.02,1) |
| | 1.00 | (250.26,246.22,175) | (152.86,148.38,107) | (59.93,54.52,43) | (26.43,21.11,20) | (14.64,9.77,12) | (9.70,5.52,9) | (4.99,2.35,5) | (3.20,1.39,3) | (2.29,0.94,2) | (1.78,0.70,2) | (1.23,0.43,1) | (1.00,0.04,1) |
| | 1.10 | (64.24,60.57,46) | (52.00,48.17,37) | (32.08,28.04,24) | (19.00,14.96,15) | (12.25,8.47,10) | (8.67,5.30,8) | (4.73,2.42,4) | (3.11,1.45,3) | (2.27,0.99,2) | (1.77,0.74,2) | (1.24,0.45,1) | (1.00,0.06,1) |
| | 1.25 | (15.73,12.67,12) | (14.82,11.79,12) | (12.69,9.79,10) | (10.23,7.47,8) | (8.13,5.55,7) | (6.53,4.16,6) | (4.12,2.28,4) | (2.89,1.46,3) | (2.18,1.03,2) | (1.74,0.77,2) | (1.26,0.46,1) | (1.01,0.08,1) |
| | 1.50 | (5.40,3.58,5) | (5.32,3.52,5) | (5.10,3.33,4) | (4.76,3.06,4) | (4.37,2.76,4) | (3.97,2.45,3) | (3.08,1.78,3) | (2.43,1.30,2) | (1.97,0.99,2) | (1.65,0.77,1) | (1.26,0.48,1) | (1.02,0.13,1) |
| | 1.75 | (3.15,1.93,3) | (3.13,1.91,3) | (3.07,1.87,3) | (2.98,1.80,3) | (2.86,1.71,2) | (2.73,1.61,2) | (2.37,1.34,2) | (2.03,1.09,2) | (1.75,0.89,2) | (1.54,0.72,1) | (1.25,0.48,1) | (1.03,0.16,1) |
| | 2.00 | (2.25,1.30,2) | (2.24,1.29,2) | (2.22,1.27,2) | (2.19,1.25,2) | (2.14,1.21,2) | (2.08,1.17,2) | (1.91,1.04,2) | (1.74,0.90,2) | (1.57,0.77,1) | (1.43,0.65,1) | (1.22,0.46,1) | (1.03,0.18,1) |
| SS-$\chi^2$ | 0.25 | (2.75,0.66,3) | (2.74,0.66,3) | (2.71,0.65,3) | (2.64,0.64,3) | (2.52,0.60,3) | (2.39,0.56,2) | (2.07,0.42,2) | (1.85,0.37,2) | (1.56,0.50,2) | (1.14,0.35,1) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.50 | (6.90,2.26,7) | (6.83,2.25,7) | (6.58,2.20,6) | (6.05,2.02,6) | (5.35,1.70,5) | (4.67,1.40,5) | (3.39,0.93,3) | (2.52,0.66,2) | (1.98,0.48,2) | (1.60,0.50,2) | (1.04,0.20,1) | (1.00,0.00,1) |
| | 0.75 | (36.69,28.26,28) | (34.61,26.53,27) | (27.96,20.88,22) | (18.56,12.39,15) | (12.02,6.50,11) | (8.58,3.83,8) | (4.76,1.76,5) | (3.12,1.07,3) | (2.26,0.74,2) | (1.75,0.59,2) | (1.16,0.37,1) | (1.00,0.01,1) |
| | 0.90 | (260.25,254.03,183) | (179.82,173.79,127) | (72.62,66.31,52) | (29.33,22.99,22) | (15.45,9.70,13) | (10.07,5.22,9) | (5.13,2.20,5) | (3.26,1.30,3) | (2.33,0.88,2) | (1.79,0.67,2) | (1.22,0.42,1) | (1.00,0.02,1) |
| | 1.00 | (249.86,245.77,174) | (155.47,150.72,109) | (61.33,55.90,44) | (26.98,21.51,21) | (14.94,9.93,13) | (9.89,5.60,9) | (5.08,2.39,5) | (3.25,1.41,3) | (2.33,0.95,2) | (1.80,0.71,2) | (1.24,0.44,1) | (1.00,0.04,1) |
| | 1.10 | (70.55,66.72,50) | (56.35,52.39,40) | (33.98,29.80,25) | (19.79,15.58,15) | (12.62,8.70,11) | (8.90,5.41,8) | (4.83,2.46,4) | (3.17,1.48,3) | (2.31,1.01,2) | (1.80,0.75,2) | (1.26,0.45,1) | (1.00,0.06,1) |
| | 1.25 | (16.78,13.54,13) | (15.78,12.57,12) | (13.37,10.32,11) | (10.72,7.82,9) | (8.43,5.74,7) | (6.73,4.26,6) | (4.22,2.32,4) | (2.94,1.49,3) | (2.22,1.04,2) | (1.76,0.78,2) | (1.27,0.47,1) | (1.01,0.09,1) |
| | 1.50 | (5.65,3.71,5) | (5.56,3.63,5) | (5.31,3.45,5) | (4.95,3.17,4) | (4.53,2.84,4) | (4.11,2.52,4) | (3.16,1.81,3) | (2.48,1.33,2) | (2.01,1.01,2) | (1.67,0.78,2) | (1.27,0.49,1) | (1.02,0.13,1) |
| | 1.75 | (3.27,1.98,3) | (3.24,1.97,3) | (3.18,1.92,3) | (3.08,1.85,3) | (2.96,1.76,3) | (2.81,1.65,2) | (2.43,1.37,2) | (2.08,1.11,2) | (1.79,0.90,2) | (1.56,0.73,1) | (1.26,0.49,1) | (1.03,0.17,1) |
| | 2.00 | (2.32,1.33,2) | (2.31,1.33,2) | (2.29,1.31,2) | (2.25,1.28,2) | (2.20,1.24,2) | (2.14,1.20,2) | (1.96,1.07,2) | (1.77,0.92,2) | (1.60,0.79,1) | (1.45,0.67,1) | (1.23,0.47,1) | (1.04,0.19,1) |

Table B4. (ARL, SDRL, MDRL) values of the six proposed CAM charts for different shifts in the process mean ($\delta$) and variance ($\theta$). $ARL_0 \cong 250$ and $k_2 = 1$.

| Chart | $\theta$ | $\delta$ | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 0.00 | 0.10 | 0.20 | 0.30 | 0.40 | 0.50 | 0.75 | 1.00 | 1.25 | 1.5 | 2.00 | 3.00 |
| Max-3T | 0.25 | (3.40,0.50,3) | (3.40,0.50,3) | (3.40,0.50,3) | (3.40,0.50,3) | (3.40,0.50,3) | (3.40,0.50,3) | (3.39,0.49,3) | (2.80,0.42,3) | (2.00,0.14,2) | (1.52,0.50,2) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.50 | (6.57,1.93,6) | (6.57,1.93,6) | (6.58,1.94,6) | (6.57,1.93,6) | (6.53,1.87,6) | (6.29,1.63,6) | (4.43,1.08,4) | (2.81,0.73,3) | (1.99,0.50,2) | (1.51,0.51,2) | (1.01,0.11,1) | (1.00,0.00,1) |
| | 0.75 | (65.07,59.29,47) | (64.42,58.51,46) | (58.39,52.55,42) | (38.29,32.72,28) | (19.28,14.33,15) | (10.69,6.38,9) | (4.60,1.84,4) | (2.84,1.04,3) | (2.01,0.71,2) | (1.54,0.56,2) | (1.07,0.25,1) | (1.00,0.00,1) |
| | 0.90 | (542.59,538.71,377) | (350.96,347.32,244) | (123.86,119.65,87) | (42.71,38.66,31) | (18.78,15.02,14) | (10.47,7.07,9) | (4.58,2.18,4) | (2.85,1.21,3) | (2.03,0.82,2) | (1.57,0.61,2) | (1.11,0.31,1) | (1.00,0.01,1) |
| | 1.00 | (250.38,247.38,175) | (168.39,165.46,118) | (73.03,69.83,52) | (31.85,28.60,23) | (16.28,13.13,12) | (9.84,6.91,8) | (4.52,2.36,4) | (2.85,1.31,3) | (2.05,0.88,2) | (1.59,0.65,2) | (1.13,0.35,1) | (1.00,0.02,1) |
| | 1.10 | (66.98,64.59,47) | (56.44,53.90,40) | (36.76,34.13,26) | (21.72,19.01,16) | (13.34,10.67,10) | (8.90,6.40,7) | (4.42,2.48,4) | (2.84,1.40,3) | (2.06,0.94,2) | (1.61,0.69,2) | (1.16,0.37,1) | (1.00,0.03,1) |
| | 1.25 | (16.97,14.78,13) | (16.11,13.91,12) | (13.91,11.72,10) | (11.24,9.08,9) | (8.80,6.70,7) | (6.88,4.88,6) | (4.08,2.44,4) | (2.77,1.47,2) | (2.05,1.01,2) | (1.63,0.73,1) | (1.19,0.41,1) | (1.00,0.06,1) |
| | 1.50 | (5.49,3.90,5) | (5.41,3.83,4) | (5.21,3.65,4) | (4.90,3.37,4) | (4.53,3.04,4) | (4.13,2.70,4) | (3.18,1.92,3) | (2.47,1.38,2) | (1.97,1.02,2) | (1.63,0.78,1) | (1.23,0.46,1) | (1.01,0.11,1) |
| | 1.75 | (3.16,1.99,3) | (3.14,1.97,3) | (3.09,1.94,3) | (3.01,1.87,3) | (2.91,1.79,3) | (2.78,1.69,2) | (2.43,1.42,2) | (2.09,1.16,2) | (1.79,0.93,2) | (1.56,0.75,1) | (1.25,0.48,1) | (1.02,0.15,1) |
| | 2.00 | (3.40,0.50,3) | (3.40,0.50,3) | (3.40,0.50,3) | (3.40,0.50,3) | (3.40,0.50,3) | (3.40,0.50,3) | (3.39,0.49,3) | (2.80,0.42,3) | (2.00,0.14,2) | (1.52,0.50,2) | (1.00,0.00,1) | (1.00,0.00,1) |
| Max-4T | 0.25 | (2.94,0.38,3) | (2.94,0.38,3) | (2.94,0.38,3) | (2.94,0.38,3) | (2.94,0.38,3) | (2.94,0.38,3) | (2.94,0.37,3) | (2.72,0.45,3) | (2.00,0.14,2) | (1.52,0.50,2) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.50 | (5.98,1.97,6) | (5.98,1.97,6) | (5.98,1.98,6) | (5.97,1.97,6) | (5.95,1.92,6) | (5.78,1.71,6) | (4.29,1.05,4) | (2.80,0.72,3) | (2.00,0.50,2) | (1.51,0.51,2) | (1.01,0.12,1) | (1.00,0.00,1) |
| | 0.75 | (52.59,47.48,38) | (52.25,47.08,38) | (48.41,43.22,35) | (34.02,28.84,25) | (18.37,13.53,14) | (10.51,6.24,9) | (4.61,1.83,4) | (2.85,1.04,3) | (2.02,0.71,2) | (1.54,0.56,2) | (1.07,0.26,1) | |
| | 0.90 | (394.89,390.95,275) | (283.62,279.27,198) | (115.35,111.26,81) | (41.91,37.90,30) | (18.65,14.87,14) | (10.48,7.07,9) | (4.59,2.18,4) | (2.86,1.21,3) | (2.03,0.82,2) | (1.57,0.61,2) | (1.11,0.31,1) | (1.00,0.01,1) |
| | 1.00 | (250.39,247.90,174) | (168.90,166.25,118) | (73.44,70.16,52) | (31.94,28.66,23) | (16.32,13.15,12) | (9.86,6.92,8) | (4.54,2.36,4) | (2.86,1.31,3) | (2.05,0.88,2) | (1.59,0.65,2) | (1.14,0.35,1) | (1.00,0.02,1) |
| | 1.10 | (71.07,68.93,50) | (59.38,56.97,42) | (37.97,35.33,27) | (22.15,19.44,16) | (13.47,10.80,10) | (8.94,6.42,7) | (4.43,2.48,4) | (2.84,1.40,3) | (2.06,0.94,2) | (1.61,0.69,2) | (1.16,0.38,1) | (1.00,0.03,1) |
| | 1.25 | (17.74,15.56,13) | (16.78,14.62,12) | (14.39,12.21,11) | (11.50,9.34,9) | (8.95,6.85,7) | (6.96,4.97,6) | (4.11,2.46,4) | (2.77,1.48,2) | (2.06,1.01,2) | (1.63,0.74,1) | (1.19,0.41,1) | (1.00,0.06,1) |
| | 1.50 | (5.59,4.02,5) | (5.51,3.95,5) | (5.30,3.76,4) | (4.98,3.47,4) | (4.59,3.13,4) | (4.17,2.77,4) | (3.20,1.96,3) | (2.47,1.40,2) | (1.97,1.03,2) | (1.62,0.78,1) | (1.23,0.46,1) | (1.01,0.11,1) |
| | 1.75 | (3.18,2.04,3) | (3.15,2.02,3) | (3.10,1.98,3) | (3.02,1.92,3) | (2.92,1.82,2) | (2.79,1.72,2) | (2.43,1.44,2) | (2.09,1.17,2) | (1.79,0.94,2) | (1.55,0.75,1) | (1.25,0.48,1) | (1.02,0.15,1) |
| | 2.00 | (2.25,1.34,2) | (2.25,1.34,2) | (2.23,1.32,2) | (2.20,1.30,2) | (2.15,1.26,2) | (2.10,1.22,2) | (1.94,1.09,2) | (1.77,0.95,2) | (1.60,0.82,1) | (1.45,0.69,1) | (1.23,0.48,1) | (1.03,0.18,1) |
| Max-$\chi^2$ | 0.25 | (2.27,0.59,2) | (2.27,0.59,2) | (2.27,0.59,2) | (2.27,0.59,2) | (2.27,0.59,2) | (2.27,0.59,2) | (2.27,0.59,2) | (2.21,0.54,2) | (1.94,0.27,2) | (1.56,0.50,2) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.50 | (5.71,2.21,5) | (5.71,2.22,5) | (5.71,2.22,5) | (5.70,2.21,5) | (5.68,2.17,5) | (5.53,1.96,5) | (4.19,1.15,4) | (2.82,0.73,3) | (2.03,0.50,2) | (1.55,0.51,2) | (1.02,0.13,1) | (1.00,0.00,1) |
| | 0.75 | (47.84,43.42,35) | (47.60,43.10,34) | (44.75,40.24,32) | (32.76,28.02,24) | (18.26,13.57,14) | (10.59,6.30,9) | (4.69,1.85,4) | (2.90,1.05,3) | (2.05,0.72,2) | (1.57,0.57,2) | (1.08,0.27,1) | (1.00,0.00,1) |
| | 0.90 | (283.43,280.02,198) | (227.02,223.80,158) | (109.37,105.39,77) | (42.36,38.31,31) | (19.05,15.20,14) | (10.70,7.19,9) | (4.68,2.21,4) | (2.91,1.22,3) | (2.07,0.83,2) | (1.60,0.62,2) | (1.12,0.33,1) | (1.00,0.01,1) |
| | 1.00 | (250.43,248.13,174) | (173.45,171.01,121) | (77.07,73.73,54) | (33.35,29.98,24) | (16.89,13.61,13) | (10.14,7.09,8) | (4.64,2.40,4) | (2.91,1.33,3) | (2.09,0.90,2) | (1.61,0.66,2) | (1.15,0.36,1) | (1.00,0.02,1) |
| | 1.10 | (80.51,78.00,57) | (66.61,64.14,47) | (41.36,38.64,30) | (23.58,20.74,17) | (14.11,11.34,11) | (9.25,6.64,8) | (4.54,2.53,4) | (2.90,1.42,3) | (2.10,0.95,2) | (1.64,0.70,2) | (1.17,0.39,1) | (1.00,0.04,1) |
| | 1.25 | (19.43,17.12,14) | (18.36,16.06,14) | (15.61,13.33,12) | (12.32,10.04,9) | (9.45,7.26,7) | (7.28,5.18,6) | (4.23,2.53,4) | (2.84,1.51,3) | (2.10,1.03,2) | (1.66,0.75,2) | (1.21,0.42,1) | (1.00,0.06,1) |
| | 1.50 | (5.90,4.24,5) | (5.83,4.18,5) | (5.59,3.96,5) | (5.24,3.64,4) | (4.81,3.27,4) | (4.35,2.88,4) | (3.31,2.02,3) | (2.54,1.43,2) | (2.01,1.05,2) | (1.65,0.80,1) | (1.25,0.47,1) | (1.01,0.12,1) |
| | 1.75 | (3.31,2.12,3) | (3.29,2.10,3) | (3.23,2.06,3) | (3.15,1.98,3) | (3.03,1.89,3) | (2.89,1.79,2) | (2.51,1.49,2) | (2.14,1.21,2) | (1.83,0.97,2) | (1.58,0.77,1) | (1.26,0.50,1) | (1.03,0.16,1) |
| | 2.00 | (2.33,1.38,2) | (2.32,1.38,2) | (2.30,1.36,2) | (2.27,1.34,2) | (2.23,1.30,2) | (2.17,1.26,2) | (2.00,1.13,2) | (1.82,0.98,2) | (1.64,0.84,1) | (1.48,0.71,1) | (1.24,0.49,1) | (1.04,0.19,1) |
| SS-3T | 0.25 | (4.12,0.33,4) | (4.11,0.33,4) | (4.09,0.32,4) | (4.01,0.36,4) | (3.74,0.50,4) | (3.28,0.46,3) | (2.65,0.48,3) | (2.02,0.12,2) | (1.92,0.27,2) | (1.25,0.43,1) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.50 | (7.77,2.09,7) | (7.71,2.09,7) | (7.52,2.09,7) | (7.06,2.08,7) | (6.20,1.92,6) | (5.19,1.55,5) | (3.47,0.86,3) | (2.52,0.61,2) | (1.96,0.44,2) | (1.56,0.50,2) | (1.02,0.15,1) | (1.00,0.00,1) |
| | 0.75 | (83.52,76.60,60) | (77.42,70.86,56) | (58.35,52.57,42) | (33.22,28.26,25) | (17.02,12.54,13) | (9.96,5.91,8) | (4.66,1.82,4) | (2.99,1.04,3) | (2.16,0.72,2) | (1.67,0.58,2) | (1.12,0.33,1) | (1.00,0.00,1) |
| | 0.90 | (560.60,554.38,390) | (346.86,342.07,242) | (122.98,118.66,86) | (43.58,39.16,32) | (19.41,15.31,15) | (11.00,7.24,9) | (4.92,2.25,5) | (3.09,1.26,3) | (2.21,0.85,2) | (1.71,0.65,2) | (1.17,0.38,1) | (1.00,0.02,1) |
| | 1.00 | (250.35,247.44,174) | (167.98,164.71,117) | (73.96,70.37,52) | (32.62,29.06,24) | (16.87,13.41,13) | (10.31,7.07,8) | (4.83,2.44,4) | (3.07,1.37,3) | (2.21,0.92,2) | (1.71,0.69,2) | (1.20,0.40,1) | (1.00,0.03,1) |
| | 1.10 | (63.33,60.75,45) | (53.06,50.48,38) | (34.49,31.73,25) | (20.78,17.96,15) | (13.06,10.27,10) | (8.86,6.23,7) | (4.57,2.49,4) | (2.98,1.43,3) | (2.18,0.97,2) | (1.71,0.72,2) | (1.21,0.42,1) | (1.00,0.05,1) |
| | 1.25 | (15.67,13.42,12) | (14.84,12.60,11) | (12.71,10.52,10) | (10.26,8.13,8) | (8.10,6.05,6) | (6.43,4.48,5) | (3.96,2.31,3) | (2.77,1.43,3) | (2.10,1.00,2) | (1.68,0.75,2) | (1.23,0.44,1) | (1.01,0.08,1) |
| | 1.50 | (5.19,3.59,4) | (5.12,3.53,4) | (4.90,3.34,4) | (4.57,3.07,4) | (4.20,2.76,4) | (3.83,2.44,3) | (2.96,1.74,3) | (2.34,1.27,2) | (1.90,0.95,2) | (1.60,0.74,1) | (1.24,0.46,1) | (1.01,0.12,1) |
| | 1.75 | (3.03,1.86,3) | (3.01,1.84,3) | (2.96,1.80,3) | (2.87,1.73,2) | (2.76,1.65,2) | (2.63,1.55,2) | (2.29,1.29,2) | (1.97,1.05,2) | (1.71,0.85,2) | (1.50,0.70,1) | (1.22,0.46,1) | (1.02,0.15,1) |
| | 2.00 | (2.19,1.24,2) | (2.18,1.23,2) | (2.16,1.22,2) | (2.13,1.19,2) | (2.09,1.16,2) | (2.03,1.12,2) | (1.87,1.00,2) | (1.70,0.87,1) | (1.54,0.74,1) | (1.40,0.63,1) | (1.20,0.44,1) | (1.03,0.17,1) |
| SS-4T | 0.25 | (3.37,0.49,3) | (3.36,0.49,3) | (3.32,0.48,3) | (3.26,0.45,3) | (3.13,0.40,3) | (2.92,0.40,3) | (2.23,0.42,2) | (2.00,0.08,2) | (1.77,0.42,2) | (1.13,0.33,1) | (1.00,0.00,1) | (1.00,0.00,1) |

| | | $\delta$ | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Chart | $\theta$ | 0.00 | 0.10 | 0.20 | 0.30 | 0.40 | 0.50 | 0.75 | 1.00 | 1.25 | 1.5 | 2.00 | 3.00 |
| Max-3T | 0.25 | (3.40,0.50,3) | (3.40,0.50,3) | (3.40,0.50,3) | (3.40,0.50,3) | (3.40,0.50,3) | (3.40,0.50,3) | (3.39,0.49,3) | (2.80,0.42,3) | (2.00,0.14,2) | (1.52,0.50,2) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.50 | (7.07,2.14,7) | (7.02,2.14,7) | (6.85,2.13,7) | (6.45,2.09,6) | (5.73,1.90,5) | (4.87,1.54,5) | (3.34,0.88,3) | (2.46,0.61,2) | (1.93,0.46,2) | (1.53,0.50,2) | (1.02,0.15,1) | (1.00,0.00,1) |
| | 0.75 | (66.95,60.84,48) | (62.60,56.63,45) | (48.87,43.54,36) | (29.71,24.99,22) | (16.12,11.83,13) | (9.68,5.74,8) | (4.62,1.81,4) | (2.98,1.04,3) | (2.15,0.72,2) | (1.67,0.58,2) | (1.12,0.33,1) | (1.00,0.00,1) |
| | 0.90 | (421.97,417.17,294) | (283.26,279.59,198) | (111.77,107.61,79) | (41.86,37.57,30) | (19.08,15.04,15) | (10.90,7.18,9) | (4.91,2.26,5) | (3.10,1.26,3) | (2.21,0.85,2) | (1.71,0.65,2) | (1.17,0.38,1) | (1.00,0.02,1) |
| | 1.00 | (249.45,246.82,173) | (168.01,164.91,117) | (73.95,70.56,52) | (32.74,29.12,24) | (16.92,13.42,13) | (10.34,7.10,8) | (4.85,2.44,4) | (3.08,1.37,3) | (2.22,0.92,2) | (1.72,0.69,2) | (1.20,0.41,1) | (1.00,0.03,1) |
| | 1.10 | (67.19,64.66,47) | (55.84,53.21,40) | (35.80,33.01,26) | (21.30,18.43,16) | (13.30,10.49,10) | (8.99,6.34,7) | (4.60,2.50,4) | (3.00,1.44,3) | (2.19,0.97,2) | (1.72,0.72,2) | (1.22,0.42,1) | (1.00,0.03,1) |
| | 1.25 | (16.37,14.13,12) | (15.44,13.20,12) | (13.17,10.99,10) | (10.56,8.42,8) | (8.28,6.23,7) | (6.55,4.57,5) | (4.01,2.35,4) | (2.79,1.44,3) | (2.11,1.00,2) | (1.69,0.75,2) | (1.23,0.45,1) | (1.01,0.08,1) |
| | 1.50 | (5.28,3.71,4) | (5.20,3.63,4) | (4.97,3.45,4) | (4.64,3.16,4) | (4.27,2.84,4) | (3.87,2.50,3) | (2.99,1.78,3) | (2.36,1.29,2) | (1.91,0.96,2) | (1.61,0.75,1) | (1.24,0.46,1) | (1.01,0.12,1) |
| | 1.75 | (3.05,1.91,3) | (3.03,1.89,3) | (2.97,1.84,3) | (2.88,1.77,2) | (2.77,1.69,2) | (2.64,1.59,2) | (2.29,1.32,2) | (1.98,1.07,2) | (1.71,0.86,2) | (1.50,0.70,1) | (1.23,0.46,1) | (1.02,0.15,1) |
| | 2.00 | (2.18,1.26,2) | (2.17,1.26,2) | (2.16,1.24,2) | (2.12,1.22,2) | (2.08,1.18,2) | (2.02,1.14,2) | (1.86,1.01,2) | (1.70,0.88,1) | (1.54,0.75,1) | (1.40,0.64,1) | (1.20,0.44,1) | (1.03,0.17,1) |
| SS-$\chi^2$ | 0.25 | (2.66,0.65,3) | (2.65,0.65,3) | (2.63,0.65,3) | (2.57,0.64,3) | (2.47,0.61,2) | (2.33,0.56,2) | (2.01,0.40,2) | (1.80,0.41,2) | (1.47,0.50,1) | (1.09,0.29,1) | (1.00,0.00,1) | (1.00,0.00,1) |
| | 0.50 | (6.72,2.40,6) | (6.68,2.40,6) | (6.53,2.38,6) | (6.16,2.30,6) | (5.51,2.06,5) | (4.73,1.66,5) | (3.30,0.96,3) | (2.45,0.66,2) | (1.92,0.49,2) | (1.54,0.50,2) | (1.03,0.17,1) | (1.00,0.00,1) |
| | 0.75 | (60.88,55.53,44) | (57.24,52.05,41) | (45.72,40.92,33) | (28.86,24.41,21) | (16.01,11.82,12) | (9.74,5.80,8) | (4.69,1.86,4) | (3.02,1.07,3) | (2.19,0.74,2) | (1.70,0.58,2) | (1.14,0.34,1) | (1.00,0.01,1) |
| | 0.90 | (321.99,318.10,225) | (235.39,231.57,164) | (105.51,101.53,74) | (41.66,37.44,30) | (19.26,15.17,15) | (11.08,7.28,9) | (5.00,2.29,5) | (3.15,1.28,3) | (2.26,0.87,2) | (1.74,0.65,2) | (1.19,0.39,1) | (1.00,0.02,1) |
| | 1.00 | (249.66,246.98,174) | (171.28,168.25,120) | (76.60,73.20,54) | (33.85,30.12,25) | (17.42,13.86,13) | (10.59,7.25,9) | (4.96,2.49,5) | (3.14,1.39,3) | (2.26,0.94,2) | (1.75,0.70,2) | (1.21,0.42,1) | (1.00,0.04,1) |
| | 1.10 | (74.92,72.20,53) | (61.77,58.99,44) | (38.60,35.73,28) | (22.51,19.55,17) | (13.83,10.91,11) | (9.28,6.51,8) | (4.71,2.55,4) | (3.07,1.46,3) | (2.23,0.99,2) | (1.75,0.73,2) | (1.23,0.44,1) | (1.00,0.05,1) |
| | 1.25 | (17.75,15.37,13) | (16.70,14.37,12) | (14.10,11.80,11) | (11.16,8.96,9) | (8.67,6.51,7) | (6.81,4.75,6) | (4.12,2.39,4) | (2.85,1.47,3) | (2.15,1.02,2) | (1.72,0.76,2) | (1.25,0.45,1) | (1.01,0.08,1) |
| | 1.50 | (5.54,3.88,5) | (5.46,3.81,5) | (5.21,3.60,4) | (4.85,3.30,4) | (4.44,2.94,4) | (4.02,2.59,3) | (3.08,1.82,3) | (2.41,1.31,2) | (1.95,0.98,2) | (1.63,0.76,1) | (1.25,0.47,1) | (1.02,0.12,1) |
| | 1.75 | (3.16,1.96,3) | (3.14,1.95,3) | (3.08,1.90,3) | (2.98,1.83,3) | (2.86,1.74,2) | (2.73,1.63,2) | (2.36,1.35,2) | (2.02,1.09,2) | (1.74,0.88,2) | (1.53,0.72,1) | (1.24,0.47,1) | (1.03,0.16,1) |
| | 2.00 | (2.25,1.30,2) | (2.25,1.29,2) | (2.22,1.28,2) | (2.18,1.25,2) | (2.14,1.21,2) | (2.08,1.17,2) | (1.91,1.04,2) | (1.73,0.90,2) | (1.57,0.77,1) | (1.42,0.65,1) | (1.21,0.45,1) | (1.03,0.18,1) |